

TAUminator:

Compact Neural Networks for Tau Triggering

Max Golubkov
Blavatnik School of Computer Science and AI
Tel Aviv University
maximg@mail.tau.ac.il

Tamar Averbuch
Blavatnik School of Computer Science and AI
Tel Aviv University
averbuch2@mail.tau.ac.il

1 Introduction

Identifying tau particles is important in studies of particles such as the Higgs and Z bosons. Many taus decay into a narrow spray of particles, which leaves a compact energy deposit in the ATLAS calorimeter. Ordinary jets of particles can leave a similar deposit, especially at low energy. A simple energy threshold therefore either accepts too much background or rejects useful tau candidates.

The decision must be made while collision data are arriving. The Large Hadron Collider (LHC) produces proton collisions at $40\,\mathrm{MHz}$, far faster than ATLAS can store them. Its first-stage, or Level-1, trigger therefore keeps only a small fraction for later reconstruction [1, 2]. We study compact neural networks for the calorimeter Trigger Objects (TOBs) supplied by the electron Feature Extractor (eFEX). The comparison is made at a fixed event fake rate of 0.5%: a useful model must retain more signal without accepting more background events.

We first study models that score one TOB at a time. We test which detector inputs carry useful information, whether compact convolutional networks improve on fully connected networks, and how the classifier and loss can protect the required efficiency across the energy range. We then ask a different question: can the network make a better event decision if it sees two TOBs together? The pair model applies the same small network to both TOBs, scores every pair in the event and uses the highest pair score. During training, every event receives equal weight even when it contains many pairs.

For the single-TOB study, the strongest global model accepts 69.61% of truth-τ objects, compared with 57.37% for the eFEX baseline. The pair study uses a different measure: the fraction of signal events that fire. On its final test, this efficiency rises from 34.30% for the second-highest- $p_{\mathrm{T}}^{\mathrm{TOB}}$ baseline to 46.70%, a gain of 12.40 percentage points. The three models reach event fake rates of 0.5183–0.5223%, slightly above the 0.5% limit. The efficiency improvement is therefore clear, but the final success criterion is not met. The two studies use different denominators and their percentages should not be compared directly.

Code and demo: https://github.com/Maxgolu/Tau-Trigger-NN

2 Background and Related Work

Why the trigger must filter, and why at a fixed rate. The LHC collision rate is $40\,\mathrm{MHz}$, far beyond readout capacity. The FPGA-based Level-1 trigger selects about $100\,\mathrm{kHz}$ within a few microseconds [1], using fast detector inputs; eFEX supplies calorimeter trigger objects (TOBs) [2]. Our di-tau selection is assigned a 0.5% background-event acceptance budget. The relevant comparison is therefore *signal efficiency at the same background rate*, not threshold-free ROC-AUC (Section 3; trigger context in Appendix A.1).

The tau particle and why it is hard to trigger on. A tau is a short-lived relative of the electron, produced in processes such as Higgs and Z decays. It decays before reaching the detector. We target its hadronic decays (about

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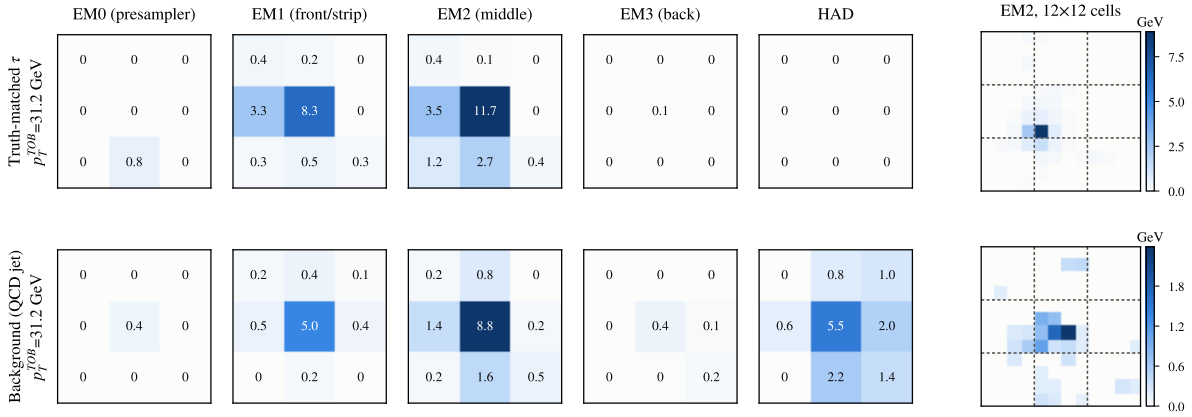


Figure 1: Calorimeter input of two typical training objects with equal $p_{\mathrm{T}}^{\mathrm{TOB}} = 31.2\,\mathrm{GeV}$. Each is chosen at its class median of $\mathrm{Frac}_{\mathrm{EM2}}$ (Equation (3)) and of deposited energy per unit $p_{\mathrm{T}}^{\mathrm{TOB}}$. Left: the five 3×3 core layers (GeV). Right: EM2 at 12×12 granularity; dashed lines mark the towers. The one-prong τ ($p_{\mathrm{T}}^{\mathrm{true}} = 42.7\,\mathrm{GeV}$) is compact, lies in EM1–EM2 and has no hadronic energy. The jet is wider and deposits much of its energy in HAD.

65% [3]), whose particles leave narrow energy deposits, or *showers*, in the calorimeter. The challenge is that far more common QCD jets can produce similar deposits, especially at low energy. A loose energy cut admits too many jets; a tight cut loses low-energy taus. We therefore use shower shape as well as energy to distinguish them (Appendix A.2).

Calorimeter inputs. Each TOB has a $5 \times 3 \times 3$ core tensor spanning four electromagnetic layers (EM0–EM3) and one hadronic layer [4], plus a finer 12×12 EM2 view of the same angular region. For example, the real tau candidate with $p_{\mathrm{T}}^{\mathrm{TOB}} = 31.2\,\mathrm{GeV}$ in Figure 1 is represented by five 3×3 energy maps (45 values) and 144 EM2-bin energies, not by a photograph. Available scalars are $p_{\mathrm{T}}^{\mathrm{TOB}}$, position and an eFEX BDT score. Truth quantities supply labels, loss weights and evaluation variables, never inference inputs; Appendix A specifies their construction.

Prior work. This project continues the study of Kernan and Areva Cohen [5], carried out with physicists of the group. They designed physically motivated shower features: per-layer energy sums, core dominance and sparsity (Appendix B.2), and EM2 shape variables such as maxima, peak distances and widths (Appendix B.3). They then tested how well a $[32, 16]$ MLP trained with BCE learns from them. EM1 and EM2 carried most of the separating power, and larger MLPs did not help. Their strongest configuration is shown separately in Figure 4b because its evaluation protocol differs. On related simulations, Yaari et al. [6] compared boosted trees, small MLPs and ResNet-style CNNs for Level-1 τ selection; the ResNets did not fit the FPGA resources. Tools such as `hls4ml` [7] make small fixed-latency networks deployable.

CNNs for detector images. Compact convolution–pooling stacks can beat much larger ResNets at low false-positive rates [8]. IceCube uses separate input branches for detector regions of different granularity [9], and strided convolutions can serve as a learned down-sampling front end [10]. These ideas shaped our designs in Section 4.3.

Learning at a fixed operating point. Efficiency at a fixed fake rate is a *non-decomposable* objective, because the threshold is itself a quantile of the scores. Such objectives can be optimised with Lagrangian relaxations [11]. With non-differentiable rate constraints, one can train on smooth surrogates and enforce the constraints from the true rates [12]. Our constrained loss (Section 4.2) follows this separation.

3 Experimental Setting

Data. We use Monte Carlo τ-signal and QCD dijet-background samples from the resimulated FEX truth/all-cell production, with detector-level inputs and generator-level truth. Per event, we retain up to the six highest-$p_{\mathrm{T}}^{\mathrm{TOB}}$ tau Trigger Objects (TOBs) from the simulated Level-1 Electron Feature Extractor (eFEX), discarding events without TOBs. A signal-sample TOB is positive if its closest visible truth τ satisfies $\Delta R < 0.15$; all other TOBs are negative. The inputs described in Section 2 are binned within a 0.3×0.3 (η, ϕ) window around each TOB. The retained dataset

contains 72,404 signal and 249,022 background events, 1.81M objects and 110,921 matched τ objects. For each of three seeds, complete events are randomly split $70/10/20\%$ into training, validation and test sets; grouping by (sample, event number) prevents event leakage and identifier collisions. Sample identifiers, input construction and split statistics are given in Appendix A.

Event decision and calibration. A model assigns every object i of event e a score $s_{e,i} = f_\theta(x_{e,i})$, computed from that object's inputs $x_{e,i}$. An object is accepted when $s_{e,i} \geq t$, and the event fires when at least two objects are accepted (Figure 3b):

$$A_e(t) = \mathbf{1}\Big[\textstyle\sum_i \mathbf{1}[s_{e,i} \geq t] \geq 2\Big] = \mathbf{1}\big[s_{e,(2)} \geq t\big], \tag{1}$$

where $s_{e,(2)}$ is the second-highest object score in event e. The *event-level* fake rate $\mathrm{FPR}_{\mathrm{ev}}(t)$ is the fraction of background events $\mathcal{B}$ that fire. It is not the fraction of background objects. The threshold is the lowest score value that keeps the fake rate within the budget:

$$t^\star = \min\big\{t \in \{s_{e,(2)}\}_{e\in\mathcal{B}} : \mathrm{FPR}_{\mathrm{ev}}(t) \leq \varepsilon_{\mathrm{B}}\big\}, \qquad \varepsilon_{\mathrm{B}} = 0.005. \tag{2}$$

The search runs over the unique score values, so tied scores are never split, and deployment uses the same $\geq$ comparison. A discrete score may therefore reach only a fake rate below 0.5%, so we always report the *achieved* rate (Appendix C).

Metrics and baseline. Signal efficiency is the fraction of truth-τ objects accepted, reported versus true p_{T} (the *turn-on*) and globally. The baseline accepts objects with $p_{\mathrm{T}}^{\mathrm{TOB}} \geq C$ under the same two-object event rule. For fixed-rate comparisons, both are recalibrated on the same test background at 0.5%; achieved rates are reported, while validation-calibrated performance is a separate transfer check. The goal is earlier turn-on without sacrificing efficiency above $25\,\mathrm{GeV}$. Reporting bands, baseline response and calibration distinctions are detailed in Appendix C.1.

Hardware constraints. An indicative FPGA budget of about 4,000 8-bit weights, together with changing input and operation constraints, motivated a range of compact candidates rather than one hardware-specific optimum. We compare roughly 2k-parameter MLPs, 4–5k EM2-only CNNs and 9k dual-branch CNNs. Counts include BatchNorm; parameter count alone does not establish hardware feasibility (Appendix G.8).

Training and selection. Inputs are standardised on training data. BCE-based training uses Adam [13] (10^{-3}, batch size 256, 20 epochs); results average three seeds. We *select for the trigger, not for BCE*: each epoch is calibrated on validation background and ranked by validation efficiency or the configured trigger objective at ε_{B}. Minimum-BCE checkpoints remain references. Test data do not select features, losses, budgets, architectures or checkpoints. Sweeps, fine-tuning settings and selection details appear in Appendices F and H.

4 Methodology

To support the changing hardware requirements of Section 3, we studied four complementary axes: data and features (Section 4.1), classifier and loss (Section 4.2), architecture (Section 4.3) and the input unit (Section 4.4). Matched comparisons isolate individual mechanisms where possible. Models are named by their inputs, architecture and training objective; Table 7 (Appendix G.1) explains every name.

4.1 Data exploration and feature engineering

We compared class-normalised distributions of τ and non-τ objects on the training split, overall and within the $p_{\mathrm{T}}^{\mathrm{TOB}}$ intervals $[5, 10)$, $[10, 20)$, $[20, 40)$, $[40, 80)$ and $[80, 120)\,\mathrm{GeV}$. A variable carries information beyond energy only if the classes still differ *inside* each interval. All features live in a modular registry and are selected by configuration (Appendix B).

Energy: a strong separator and a shortcut. $p_{\mathrm{T}}^{\mathrm{TOB}}$ is the strongest single separator (Figure 2a). Non-τ objects pile up at the $5\,\mathrm{GeV}$ threshold, while τ objects have a long tail. For the same reason it is a *shortcut*: a network trained with BCE lowers its loss by learning the energy scale and has little incentive to learn the weaker, complementary shape information. This observation motivated the classifier of Section 4.2.

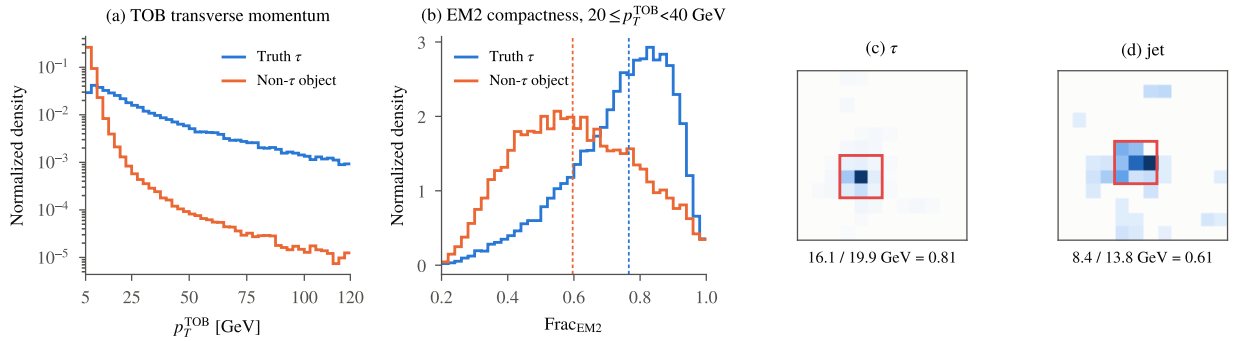


Figure 2: Feature study (training split, normalised per class). (a) $p_{\mathrm{T}}^{\mathrm{TOB}}$, log scale. (b) $\mathrm{Frac}_{\mathrm{EM2}}$ for $20 \le p_{\mathrm{T}}^{\mathrm{TOB}} < 40\,\mathrm{GeV}$; dashed lines mark the medians. (c,d) Computation of $\mathrm{Frac}_{\mathrm{EM2}}$ for the objects of Figure 1: the red square is the best 3×3 window.

EM2 compactness. The most useful engineered variable is the fraction of the 12×12 EM2 energy contained in its most energetic 3×3 cell window:

$$\mathrm{Frac}_{\mathrm{EM2}} = \max_{w \in \mathcal{W}} \sum_{(a,b) \in w} E_{ab} \Big/ \sum_{a,b} E_{ab}, \tag{3}$$

where $\mathcal{W}$ is the set of the 100 overlapping 3×3 windows (Figure 2c,d). A cell window is smaller than one tower, so $\mathrm{Frac}_{\mathrm{EM2}}$ measures compactness at a granularity the core tensor cannot resolve. Unlike $p_{\mathrm{T}}^{\mathrm{TOB}}$, its separation persists within fixed energy intervals and grows with energy (Figure 2b, Figure 6).

Inputs used below. Besides $\mathrm{Frac}_{\mathrm{EM2}}$ (F), four inputs stayed competitive at the operating point: the 45 raw core cells (CT), the strongest single input throughout; 15 per-layer summaries (SUM), which are the sum S_ℓ, the dominance $2M_\ell - S_\ell$ (M_ℓ the maximum cell) and the sparsity $\#\{\text{cells} > 0.1M_\ell\}$ of each layer [5]; the "core physics" vector (CP): BDT score, $p_{\mathrm{T}}^{\mathrm{TOB}}$, and η, ϕ relative to the leading TOB; and the EM2 image, which only the CNNs read. Event-topology variables (the $p_{\mathrm{T}}^{\mathrm{TOB}}$ sum, the second-highest $p_{\mathrm{T}}^{\mathrm{TOB}}$, the separation of the two leading TOBs) clearly separate events by τ multiplicity. However, they take the same value for every object of an event, cannot rank the objects within it, and did not improve on object-level inputs at ε_{B} (Section 4.4, Appendix B).

4.2 Classifier and loss design

BCE measures the average quality of object-level probabilities. The trigger instead depends on the far tail of the *event-level* background distribution and has explicit requirements per energy region. We addressed this mismatch in three steps, each changing one mechanism: the decision rule, the signal weighting and the training objective.

The TOB–NN OR classifier. Instead of asking a network to relearn the energy threshold, we pair it with the deterministic TOB comparator: accept an object through either branch, then require two accepted objects per event,

$$a_{e,i} = \mathbf{1}[p_{\mathrm{T},e,i}^{\mathrm{TOB}} \ge C] \vee \mathbf{1}[s_{e,i} \ge t], \qquad A_e = \mathbf{1}\Big[\textstyle\sum_i a_{e,i} \ge 2\Big]. \tag{4}$$

The TOB branch protects the high-energy plateau; the NN supplies complementary shape-based decisions. Thresholds are calibrated jointly, counting mixed-branch events exactly. Validation selects the TOB budget b to maximise the equally weighted efficiency gain across 25–100 GeV, subject to the event fake-rate limit and regional protection guards. A smaller selected $b^\star$ suggests less reliance on the comparator, not a standalone measure of NN quality. Training remains (weighted) BCE; the OR changes the decision rule only. Appendix D gives the objective, cross-fitted calibration, guards and earlier two-expert trial.

Spectrum-aware signal weighting. We reweight each object's contribution to binary cross-entropy:

$$\mathcal{L}_{\mathrm{wBCE}} = -\frac{1}{N} \sum_i w_i \big[y_i \log s_i + (1 - y_i) \log(1 - s_i)\big], \tag{5}$$

where y_i is the signal label and s_i the NN score. Background keeps $w_i = 1$; for signal, $w_i = r(t_i)/\langle r \rangle_{\mathrm{train},S}$, with truth momentum t_i in GeV. Mean-one normalisation preserves the total signal weight while redistributing its gradient

across energies. We compare two raw profiles:

$$r_{\mathrm{IF}}(t) = \mathrm{clip}\big(\bar{n}/n_{k(t)},\, 0.2,\, 5\big)\,, \quad r_p(t) = \mathrm{clip}(t, 10, 200)^{-p}. \tag{6}$$

Inverse frequency (IF) uses the training-signal count n_k in each populated 5 GeV bin over 25–100 GeV, with $\bar{n}$ their mean; outside these bins, $r_{\mathrm{IF}} = 1$. Rare bins receive larger weights, approximately balancing their total influence. *Power-law weighting* varies smoothly: $p = -1$ doubles the weight when energy doubles away from the clamp, whereas $p = 0$ recovers BCE. Since soft taus dominate the sample, both IF and negative p shift learning towards rarer, cleaner energetic examples without resampling the data. This often helped more than further emphasising ambiguous low-energy taus, but depended on the inputs. Full profiles, scans and the transfer hypothesis appear in Appendix E; comparisons are in Appendix I.4.

A constrained trigger objective. A fixed weight profile applies the same emphasis whether or not a protection is at risk. Our last step therefore optimises the trigger objective itself. It had a second, more general aim: to find out what a loss that truly represents the trigger problem must look like, and which properties it must have. We collect these conclusions in Section 6. The problem is

$$\max_{\theta} \sum_{k \in \mathcal{K}_J} w_k \big[\epsilon_k(\theta) - \epsilon_k^{\mathrm{TOB}}\big] \quad \text{s.t.} \quad \mathrm{FPR}_{\mathrm{ev}}(\theta) \le \varepsilon_{\mathrm{B}}, \;\; \epsilon_k(\theta) \ge r_k \;\; (k \in \mathcal{K}_C). \tag{7}$$

In the original MLP formulation, the regions to improve, $\mathcal{K}_J = \{[25, 32), [32, 40), [40, 60)\}$ GeV, are kept separate from the regions to protect, $\mathcal{K}_C = \mathcal{K}_J \cup \{[60, 120)\}$. The saturated region is protected but not rewarded, because there is almost no room for gain there and its measurement is dominated by noise. The CNN extension also rewards 60–100 GeV and splits the saturation guards (Appendix F). We train on the Lagrangian of (7):

$$\mathcal{L}(\theta, \lambda) = \underbrace{-\sum_{k \in \mathcal{K}_J} w_k \big[\tilde{\epsilon}_k(\theta) - \epsilon_k^{\mathrm{TOB}}\big]}_{\text{reward: turn-on gain}} + \lambda_{\mathrm{FPR}} \underbrace{\frac{\widetilde{\mathrm{FPR}}(\theta) - \varepsilon_{\mathrm{B}}}{\varepsilon_{\mathrm{B}}}}_{\text{fake-rate violation}} + \sum_{k \in \mathcal{K}_C} \lambda_k \underbrace{\big(r_k - \tilde{\epsilon}_k(\theta)\big)}_{\text{deficit in region } k}. \tag{8}$$

- The *reward* is the efficiency gain over the baseline in the turn-on regions. For the MLP, its fixed weights $w = (0.35, 0.35, 0.30)$ state our physics preference. They are not learned, because an optimiser allowed to choose them would put all the weight on the easiest region.
- The *fake-rate term* measures the violation relative to the 0.5% target. Without this normalisation, an excess of one per mille would be numerically invisible next to an efficiency deficit of the same physical importance.
- Each *region term* compares the efficiency with a floor r_k. The floor requires a small margin over the baseline and allows at most a small loss relative to the pretrained starting model.
- The multipliers $\lambda \ge 0$ act as adaptive prices. The network parameters θ descend on $\mathcal{L}$ with Adam. Each price starts at zero, rises by projected dual ascent only while its constraint is *measured* to be violated by the exact deployed rule, and falls back when it is satisfied [12]. A fixed weight profile is the special case in which these prices are guessed in advance.

The hard threshold and two-object event rule have no useful gradient, so training uses smooth approximations while constraint checks use exact decisions. Four choices stabilise this separation:

(i) **Recalibrate the proxy threshold.** A batch-wise background quantile prevents uniformly raising all scores from appearing to improve classification.
(ii) **Anneal the temperature.** A broad early sigmoid reaches more objects; cooling from 0.10 to 0.02 sharpens it towards the hard decision.
(iii) **Measure violations out of sample.** Cross-fitting on a held-out 30% of training events prevents optimistic in-sample estimates from suppressing the dual updates.
(iv) **Start from a trained model and respect uncertainty.** Fine-tune a BCE-based checkpoint, allow statistically meaningful tolerances and calibrate deployment conservatively.

Appendix F gives the equations, feasibility rules and discarded variants.

4.3 Architectures

Every model in this paper is assembled from the same parts (Figure 3a). Optional convolutional branches read the core tensor and the EM2 image, their outputs are concatenated with scalar inputs, and a $[32, 16]$ dense head produces the object score. The model names in Table 7 describe which parts are used.

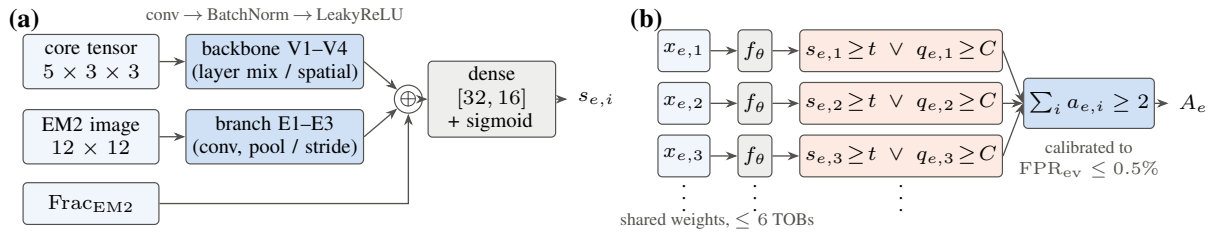


Figure 3: Overview of the single-object pipeline. (a) How every model is assembled: optional convolutional branches on the core tensor (V1–V4) and on the EM2 image (E1–E3), concatenated with scalar inputs such as $\mathrm{Frac}_{\mathrm{EM2}}$, feed a $[32, 16]$ head. The MLP uses flat inputs only. (b) How object scores become an event decision (Sections 3 and 4.4): one shared network scores every TOB; the OR classifier also accepts an object when $q_{e,i} = p_{\mathrm{T}}^{\mathrm{TOB}} \geq C$; the event fires with at least two accepted objects.

MLP. The reference model is the $[32, 16]$ ReLU MLP with a sigmoid output of [5]. On the 45 core cells plus $\mathrm{Frac}_{\mathrm{EM2}}$ (CT+F, 46 inputs) it has 2,049 parameters. Larger MLPs did not help [5], so we looked for a better inductive bias rather than more capacity.

Convolutions on the core tensor. A flat MLP must learn from data that neighbouring cells are adjacent. A convolution builds in locality and weight sharing instead. Many hand-made features are in fact fixed convolutions followed by pooling; $\mathrm{Frac}_{\mathrm{EM2}}$, for example, is an all-ones 3×3 kernel, a global maximum and a division. We tested four backbones on the $5 \times 3 \times 3$ tensor. V1 mixes the five layers at every cell ($1 \times 1 \times 5$ kernel) and learns longitudinal profiles. V2 learns local spatial patterns ($2 \times 2 \times 5$ kernel). V3 applies them in series, and V4 in parallel. Because the window is centred on the TOB seed, position matters, so the maps are flattened rather than pooled. Appendix G.4 explains each backbone step by step, and Figure 7a draws them.

A branch for the full-granularity EM2 layer. Following [8–10], the 12×12 EM2 image gets its own branch. E1 applies one 3×3 convolution with four filters and 2×2 max pooling. E2 adds a second convolution block. E3 learns its down-sampling with a stride-2 convolution. The *dual-branch* model concatenates a V-backbone, an E-branch and optionally $\mathrm{Frac}_{\mathrm{EM2}}$ into one head. Appendix G.5 and Figure 7b give the details.

Is $\mathrm{Frac}_{\mathrm{EM2}}$ still needed next to the EM2 branch? Although $\mathrm{Frac}_{\mathrm{EM2}}$ is derived from EM2, a small CNN need not learn its global normalisation easily. We therefore compare matched models with and without the scalar: does an explicit scale-free ratio complement learned local patterns? The ablation is reported in Appendix I.5; the architectural rationale is in Appendix G.6.

Towards a smaller model: EM2 only. EM2's strong contribution motivated removing the core convolution, but not the core measurements: the 45 cells or 15 layer summaries enter the dense head as scalars beside the E-branch and $\mathrm{Frac}_{\mathrm{EM2}}$. This tests whether spatial processing is worth its parameter cost (Appendices G.7 and I.5).

Stability and simplicity. Every convolution is followed by BatchNorm [14] and LeakyReLU [15], which keeps training stable on this highly imbalanced data. Inputs are standardised energies (Appendix G). Their performance is compared in Section 5.2. All parameter counts are listed in Table 8.

4.4 From individual TOBs to TOB pairs

Scoring one TOB at a time. All models above score each TOB independently, with one shared network applied to that object's inputs. The event decision is taken afterwards by the counting rule (1) or by the OR rule (4) (Figure 3b). The decision is therefore independent of TOB order. One small circuit can be replicated for every TOB slot. Training uses object labels, while calibration and evaluation are done at event level. The price is that the network sees no event context: an event-level input is identical for all objects of an event, so it cannot rank them.

Scoring two TOBs together. The single-object rule can accept two individually plausible TOBs without asking whether they form a plausible di-tau candidate. We therefore changed what the network sees. Within the at-most-six retained TOBs of each event, we keep those with finite measured $p_{\mathrm{T}}^{\mathrm{TOB}}$ and sort them by decreasing $p_{\mathrm{T}}^{\mathrm{TOB}}$; increasing `tob_index` breaks exact ties. Every unordered pair (i, j) with $i < j$ is scored once. We do not add a dummy pair. An

event with fewer than two finite TOBs therefore cannot pass through the neural branch. This keeps the $p_{\mathrm{T}}^{\mathrm{TOB}}$ baseline unchanged: the largest lower $p_{\mathrm{T}}^{\mathrm{TOB}}$ among all pairs is the second-highest finite $p_{\mathrm{T}}^{\mathrm{TOB}}$ in the event.

The same encoder is used for both TOBs. The final pair model processes both TOBs with the same weights. Each TOB contributes an unnormalised absolute-scale 12×12 EM2 image and a signed shape channel $E/\max(\sum_{ab}|E_{ab}|, 1\,\mathrm{MeV})$, together with the 45 coarse calorimeter cells, measured $p_{\mathrm{T}}^{\mathrm{TOB}}$ and $\mathrm{Frac}_{\mathrm{EM2}}$. One event-level second-highest-$p_{\mathrm{T}}^{\mathrm{TOB}}$ value is also given to the dense head. A one-convolution image branch and a 16-unit scalar branch produce a 16-value representation of each TOB. The two representations and the event value then enter a $[32, 16, 1]$ head. The complete model has $4{,}357$ trainable parameters. Sharing the encoder makes both TOBs follow the same rule, while the final head can learn whether their calorimeter patterns form a useful pair.

Every event receives equal weight during training. The training label is positive when both TOBs are labelled as tau candidates and negative when neither is. Pairs with one labelled and one unlabelled TOB are omitted from training; this label does not prove that the two TOBs match two distinct generator taus. For a positive pair, the $p = -1$ weight of (6) uses the smaller associated truth momentum; negative pairs keep unit weight. An event with many TOBs produces many more pairs. If we simply averaged the loss over pairs, such events would also receive more weight. We instead average weighted BCE over the pairs of each event and then average those event losses. A batch contains all retained pairs from up to 256 shuffled events:

$$\mathcal{L}_{\mathrm{pair}} = \frac{1}{|\mathcal{B}|}\sum_{e\in\mathcal{B}}\frac{1}{|\mathcal{P}_e|}\sum_{(i,j)\in\mathcal{P}_e} w_{e,ij}\,\ell_{\mathrm{BCE}}(s_{e,ij}, y_{e,ij}). \tag{9}$$

Thus every event contributes equally, regardless of its number of finite TOBs. Truth information is used only to define training labels and weights. During validation and testing, the model scores every pair without truth filtering.

From pair scores to one trigger decision. The neural event score is the maximum pair score, $s_e = \max_{(i,j)\in\mathcal{P}_e} s_{e,ij}$. The $p_{\mathrm{T}}^{\mathrm{TOB}}$ branch uses $q_e = \max_{(i,j)\in\mathcal{P}_e} \min(p_{\mathrm{T},e,i}^{\mathrm{TOB}}, p_{\mathrm{T},e,j}^{\mathrm{TOB}})$, and the final event decision is

$$A_e = \mathbf{1}[s_e \geq t] \vee \mathbf{1}[q_e \geq C]. \tag{10}$$

The two branches share one event-level background budget, so an event accepted by both is counted once. Checkpoints, the neural threshold and the $p_{\mathrm{T}}^{\mathrm{TOB}}$-branch budget are selected on validation independently for seeds 42, 123 and 456. The three trained models are reported separately and averaged only when the results are summarised; they are not combined into an ensemble. Exact construction, training and final-test details appear in Appendix K.

5 Results and Benchmarking

Unless stated otherwise, our results use the fixed-rate protocol of Section 3: achieved event fake rates are 0.498–0.499%, band efficiencies pool tau counts, and entries are three-seed means with standard errors. Checkpoints and OR budgets are selected on validation; test-background recalibration is used only for fixed-rate reporting. Ranking finished configurations is descriptive, not a further unbiased test. The prior-work reference has a different protocol (Appendix I.2). Table 7 decodes all model names; IF denotes inverse-frequency weighting, whereas p denotes the separate t^{-p} profile.

5.1 Benchmark: performance under different size and plateau requirements

Table 1 selects the highest-global-efficiency model satisfying each row's criterion. *All bands* means that the three-seed mean exceeds the baseline in every reported band, not that every seed does so. These are candidates for hardware evaluation, not validated FPGA implementations.

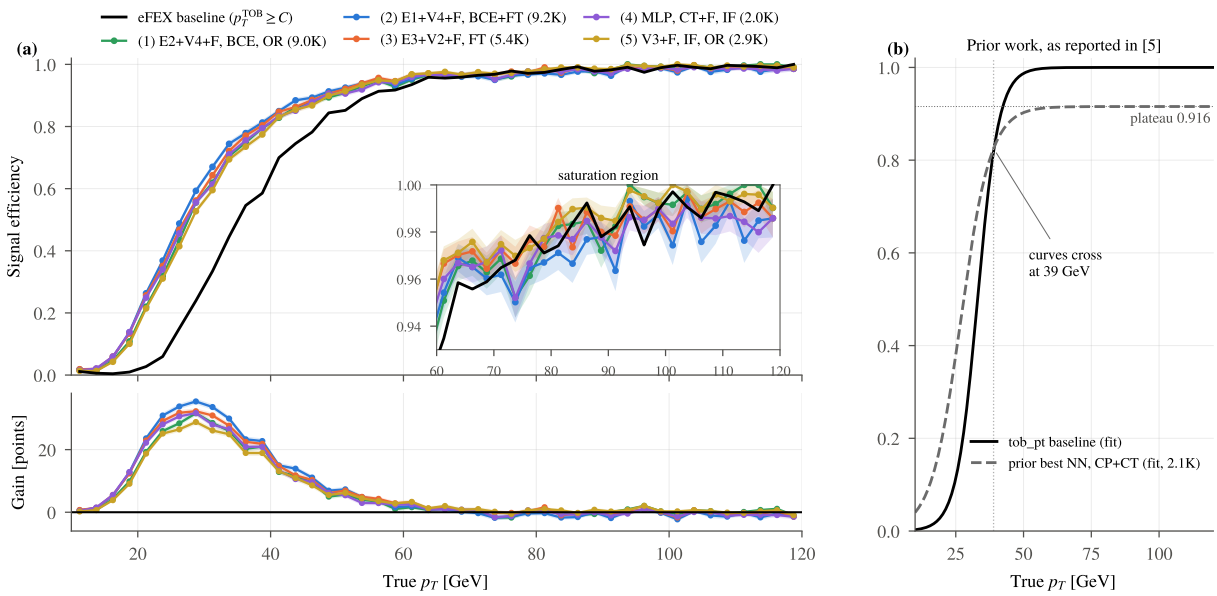


Figure 4: (a) Benchmark turn-ons; model numbers refer to Table 1. The inset shows saturation, and the lower panel shows gains over the eFEX baseline. Shading indicates statistical uncertainty. (b) Prior-work NN and baseline fits from [5], shown as a separate historical reference. The newly supplied band averages in Table 1 are not point samples of this fit (Appendix I.2). Model-name key: Table 7.

Table 1: Benchmark efficiencies [%] by true-p_T band [GeV]. Red marks our values below the baseline; counts include BatchNorm. The prior-work row is a separately evaluated reference, not part of the ranking. Model-name key: Table 7; provenance: Appendix I.2.

Criterion	Model	10–25	25–40	40–60	60–100	100–120	Global	Params
(1) all bands	E2+V4+F, BCE, OR	14.65	62.38	88.99	97.17	99.44	67.97±0.55	9.0K
(2) global	E1+V4+F, BCE+FT	17.45	66.39	90.54	96.75	98.28	69.61±0.29	9.2K
(3) global, <6K	E3+V2+F, FT	16.93	64.31	90.08	97.56	98.89	69.08±0.35	5.4K
(4) global, <3K	MLP, CT+F, IF	16.61	62.93	89.00	97.17	98.53	68.40±0.21	2.0K
(5) all bands, <3K	V3+F, IF, OR	14.13	60.42	89.14	97.89	99.50	67.47±0.22	2.9K
—	Prior best NN, CP+CT [5]	12.59	60.07	86.44	90.90	91.63	62.53	2.1K
—	eFEX baseline	2.28	35.85	81.60	96.89	99.34	57.37	—

The practical trade-off is clear: all five models turn on earlier, but the highest global efficiency, strongest plateau and smallest network do not coincide. The compact MLP retains much of the CNN gain at less than a quarter of the largest model's size. Seed-level plateau behaviour and the NN-only alternative to the compact OR model are detailed in Appendix I.3.

5.2 What the results tell us

Figure 5 highlights two controlled comparisons. The supporting tables, uncertainties and broader evidence are collected in Appendices I.4 to I.7.

(a) More weight on energetic taus can help. The results support the hypothesis in Section 4.2: emphasising energetic rather than soft taus often helped. Compared with BCE, IF reduced the selected TOB budget (Table 11). Figure 5a shows the resulting trade-off: after IF training, OR buys a small plateau gain at a larger low-energy cost, so NN-only can win globally. This is input-dependent, not a universal preference for IF or negative p (Appendix I.4).

(b) Spatial learning and engineered features complement each other. CNNs recovered some taus missed by the MLP, consistent with sensitivity to spatial shower patterns. The larger gain came from the fine-grained EM2

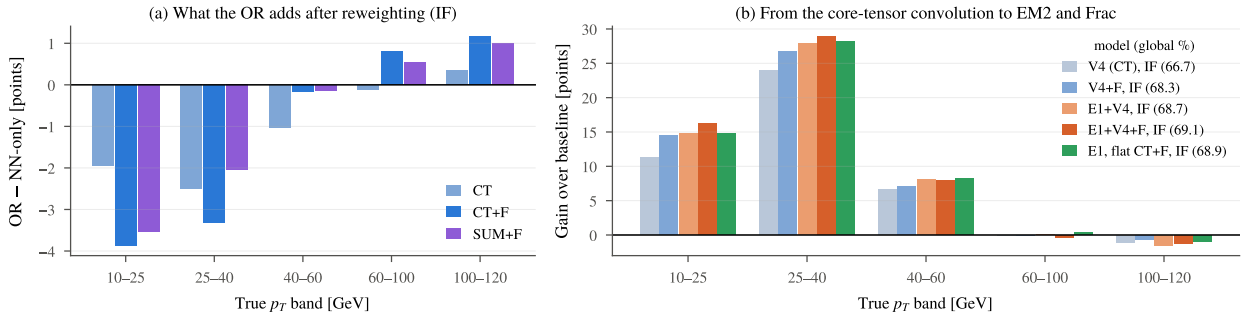


Figure 5: (a) Remaining OR trade-off after inverse-frequency (IF) weighting: adding the comparator exchanges low-energy efficiency for plateau gains. Both sides use IF; the BCE-to-IF comparison is in Table 11. (b) Effect of adding EM2 and $\mathrm{Frac}_{\mathrm{EM2}}$, then removing the core convolution. Abbreviations: Table 7. Supporting comparisons: Appendices I.4 and I.5.

input; adding $\mathrm{Frac}_{\mathrm{EM2}}$ remained useful, while removing the core convolution retained most performance with fewer parameters (Appendix I.5).

(c) More parameters were not the main route to improvement. Wider MLPs and deeper CNNs gave no systematic advantage in the tested families. Similar errors across core-only models point to limited input discrimination rather than insufficient capacity (Appendix I.6).

(d) The objective controls the trade-off across energies. Constrained fine-tuning redistributed efficiency towards protected or rewarded regions, usually with a small global change. The best loss depended on the inputs, and the lowest-energy band remained difficult across methods (Appendix I.7).

5.3 Final evaluation of the pair model

Seeing two TOBs together improves the turn-on. The selected model scores every pair of finite-$p_{\mathrm{T}}^{\mathrm{TOB}}$ TOBs and gives every event equal weight during training. Unlike the fixed-rate comparisons above, we did not recalibrate its thresholds on the test set. The checkpoint and both branch thresholds were selected on validation and then applied unchanged; the $p_{\mathrm{T}}^{\mathrm{TOB}}$ baseline uses $q_e \geq 19.4\,\mathrm{GeV}$. The final test contains 64,286 events: 14,509 signal and 49,777 background events. Its 361,685 finite TOBs form 866,872 unordered pairs; 589 events contain fewer than two finite TOBs and therefore fail both branches. The selected epochs are 17, 18 and 20 for seeds 42, 123 and 456, respectively.

For this subsection only, signal efficiency is the fraction of signal events that fire. Elsewhere in the paper it is the fraction of truth-τ objects accepted, so the two percentages are not directly comparable. For the energy-bin results, we use eligible signal events in which a pair among the four highest-$p_{\mathrm{T}}^{\mathrm{TOB}}$ TOBs has two positive training labels and two finite truth momenta. The event is assigned the lower of those two momenta. These values are therefore also not directly comparable with Table 1 and Figure 4.

Table 2: Final pair model test efficiency and achieved event fake rate. The baseline is the second-highest finite-$p_{\mathrm{T}}^{\mathrm{TOB}}$ rule. Differences are paired within the same 14,509 signal events.

Seed	Pair model [%]	Baseline [%]	Gain [points]	$\mathrm{FPR}_{\mathrm{ev}}$ [%]
42	47.07	34.30	12.77	0.5183
123	46.55	34.30	12.25	0.5223
456	46.48	34.30	12.19	0.5203
Mean	46.70	34.30	12.40 ± 0.32 SD	0.5203

Across the three models, mean signal efficiency rises from 34.30% to 46.70%, a mean gain of 12.40 percentage points with a standard deviation of 0.32 points across seeds (Table 2). The gain is concentrated in the turn-on energy ranges: the mean differences are 21.43 points for 10–25 GeV and 22.19 points for 25–60 GeV, while the already saturated region above 60 GeV gains 0.78 points. This establishes the performance of the selected pair model; the final test alone does not show which design choice caused the improvement.

The efficiency gain comes with a small rate excess. The three test event fake rates, 0.5183–0.5223%, are all above the required 0.5%. Seed 42 also misses the adjusted high-energy significance requirement ($p_{\mathrm{adj}} = 0.05946$). Our final success criterion required both conditions, so it is not met, even though every model improves the overall efficiency. The operating point should be recalibrated on a new calibration sample before deployment; the test set must not be reused for that correction. A repository-local identity audit found no overlap with the enumerated Phase-5 confirmation archives. Complete absence of every possible historical or external test exposure cannot be proven, so the result should be interpreted with that exposure uncertainty. Exact thresholds, hashes and uncertainty procedures appear in Appendix K.

6 What it TAUght us

Design the loss for the decision. Building a task-specific loss meant translating the trigger into rewards, protected regions and a fake-rate constraint, rather than merely choosing a standard classification loss. Its value was control over the efficiency profile, not a guaranteed global gain (Section 4.2 and Appendix I.7).

More gradient is not more learnability. We initially up-weighted difficult low-energy taus to improve their efficiency; this often hurt. Emphasising cleaner, higher-energy examples worked better for several inputs. Our working explanation is that useful class structure can transfer, whereas ambiguous examples may consume gradient without improving separation. Weighting must reflect what the data and model can learn, not only where we want improvement; this is a problem-dependent lesson (Appendices E and I.4).

Understand the input before enlarging the model. The earlier MLP sweep tested networks up to roughly 100 times larger without improving its small reference, while our gains came from fine-grained EM2 information and compactness features. We learned to question input discrimination before blaming capacity: a larger network cannot restore detail absent from its inputs (Section 4.1 and Appendices I.5 and I.6).

Complexity must earn its parameters. Deeper or strided EM2 branches did not systematically beat the simple E1. Local patterns and their positions may already suffice on these small images; E1's dense head also combines them globally, while further compression can lose detail. This is a plausible explanation, not proof of why a particular model lost. We therefore favour the smallest architecture that preserves the required efficiency profile (Section 4.3 and Appendix I.6).

Loss (function), but not for nothing. We learned three design requirements the hard way: (i) reward better ranking, not score inflation; (ii) give every relevant energy region an explicit reward or protection, or it can be sacrificed; and (iii) ensure gradients reach missed signal, not just objects already near the threshold. Exact out-of-sample checks and statistically meaningful tolerances complete the recipe. Appendix F.8 details all five lessons and the failures that revealed them.

Let the model see what the decision depends on. The trigger asks whether an event contains two useful TOBs, whereas the original network judged each TOB separately. Scoring TOB pairs gives the model some event context without requiring a large network over the whole event. The final test shows that the selected pair model works better, but it does not by itself show whether the gain came from pairing, the encoder or another design choice; those questions require matched validation comparisons (Sections 4.4 and 5.3).

One event should count once. An event with n TOBs produces $n(n-1)/2$ pairs. Averaging BCE over all pairs would therefore give more weight to events with more TOBs, even though the fake rate and efficiency count every event once. We first average the loss within each event and then across events. More pairs from one event are not more independent evidence (Equation (9)).

Keep the final test independent. A large efficiency gain and a missed fake-rate requirement can both be true. We therefore fixed the thresholds, denominators and success requirements before the final test. This prevents the small rate excess from being hidden by recalibration, while still reporting the useful efficiency improvement (Section 5.3 and Appendix K).

7 Tau be continued

Four directions follow from the remaining accuracy–resource trade-off:

- **Smaller CNNs.** Reduce channels and head widths, prune, quantise and distil strong models into compact students; validate the resulting efficiency, latency and FPGA resource use.
- **Adaptive loss design.** Derive target regions and tolerances from the starting model's weaknesses, with uncertainty-aware updates that adapt to the input representation.
- **Beyond pairs.** Extend the pair model to a variable-size event network whose output does not depend on TOB order. Compare it at the same event fake rate, give each event equal weight during training and use a new held-out evaluation sample.
- **Richer detector inputs.** The limited detector information available to our models leaves low-energy tau and background deposits difficult to distinguish. Richer measurements and finer spatial resolution at the input stage could therefore be crucial, rather than further model complexity alone.

Appendix J develops these directions and their evaluation requirements.

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A Detector, Data and Inputs

A.1 Trigger rate and latency context

Proton bunches cross at $40\,\mathrm{MHz}$, while front-end buffers retain detector information for only a few microseconds. Readout and storage bandwidth cannot accommodate every collision, so the Level-1 trigger makes a fixed-latency hardware decision and reduces the accepted rate to about $100\,\mathrm{kHz}$ [1]. It uses fast, reduced detector information rather than the full offline readout; the eFEX groups calorimeter deposits into candidate TOBs [2]. The accepted rate is shared among physics selections. Since ordinary background dominates, each selection's rate allocation motivates a background-acceptance constraint. The 0.5% event-level budget used here is the specification of this study, not a universal conversion from an LHC rate to a sample acceptance. Comparisons must use a common background sample and operating point.

A.2 Tau decays and the QCD background

The tau is a short-lived lepton of mass $m_\tau \simeq 1.777\,\mathrm{GeV}$ [3]; its decay products, rather than the tau itself, reach the detector. Tau pairs arise in processes including Z and Higgs decays. About 65% of tau decays are hadronic, producing a neutrino and a small number of hadrons, commonly one or three charged prongs together with neutral pions that decay into photon pairs. Electron and muon decays are handled by other triggers; throughout this paper, "tau" denotes the hadronic channel. The neutrino is invisible, while the visible hadrons and photons produce a narrow shower. QCD jets are vastly more frequent, typically wider and richer in hadronic activity, yet overlap strongly with tau showers at low energy. This combination of class imbalance and ambiguous soft showers motivates shape information in addition to energy.

A.3 Sample provenance and candidate construction

The samples are Monte Carlo simulations from the resimulated FEX truth/all-cell production, providing detector-level calorimeter information and generator-level truth. The exact production identifiers are:

- **Signal (τ events):**
  `user.dreikher.801002.Py8EG_A14NNPDF23LO_Gammatautau_MassWeight.e8557_s4422_r16130`
- **Background (QCD dijet events):**
  `user.dreikher.801165.Py8EG_A14NNPDF23LO_jj_JZ0.e8557_s4422_r16130`

The simulated eFEX, part of the Level-1 calorimeter trigger, supplies tau TOBs as candidate seeds. Within each event, TOBs are sorted by $p_{\mathrm{T}}^{\mathrm{TOB}}$ and at most six are retained; events with no TOBs are discarded. Their $p_{\mathrm{T}}^{\mathrm{TOB}}$, η and ϕ are retained as candidate-level quantities.

For each retained TOB in the signal sample, the closest visible truth τ is found using $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$, with periodic wrapping of $\Delta\phi$. The object label is positive only for $\Delta R < 0.15$. Unmatched signal-sample TOBs and all QCD-sample TOBs are negative. Thus, sample membership alone does not determine the object label. Truth information is used for labels, loss weights and offline evaluation, never as an inference input.

Calorimeter binning. Cells are selected within $|\Delta\eta| < 0.15$ and $|\Delta\phi| < 0.15$ of the TOB centre. Their transverse energies are separated into four electromagnetic layers (EM0–EM3) and one combined hadronic layer, each binned into 3×3 cells to form the $5 \times 3 \times 3$ tensor. A separate 12×12 grid represents EM2 over the *same* angular region. Transverse energies of cells assigned to the same bin are summed. The EM2 image therefore increases spatial resolution, not angular coverage; a 3×3 window within that image is not the full coarse 3×3 EM2 layer.

A.4 Calorimeter layers

Table 3 summarises the role of each layer in the core tensor. The layer order in the tensor, EM0 to HAD, follows the depth of the calorimeter. Physically, a hadronic τ decay deposits the photons of its π^0 mesons mainly in EM1–EM2, and part of the energy of its charged pions in EM2–EM3 and HAD. Its narrow opening angle keeps most of the energy in the central tower. A QCD jet carries more hadrons and more particles, so it is wider in (η, ϕ) and deeper in the calorimeter.

A.5 Coordinates and kinematic variables

The beam defines the z axis, and the plane perpendicular to it is the *transverse* plane.

Table 3: Longitudinal layers of the per-object calorimeter window.

Layer	Detector part	Physical role	Representation
EM0	presampler	energy lost in material upstream	3×3 towers
EM1	EM front (strips)	fine η segmentation; resolves $\pi^0 \to \gamma\gamma$	3×3 towers
EM2	EM middle	bulk and core of EM showers	3×3 towers, 12×12 cells
EM3	EM back	tail of energetic EM showers	3×3 towers
HAD	hadronic calorimeter	energy of charged hadrons	3×3 towers

- **Azimuth** $\phi \in [-\pi, \pi)$ is the angle around the beam in the transverse plane.
- **Pseudorapidity** $\eta = -\ln\tan(\theta/2)$ is a monotone function of the polar angle θ measured from the beam axis. $\eta = 0$ is perpendicular to the beam, and $|\eta|$ grows towards the beam. All TOBs in our data have $|\eta| < 2.5$. Equal steps in η correspond to roughly equal particle densities, which is why calorimeter towers have a fixed size of 0.1×0.1 in (η, ϕ).
- **Transverse momentum** $p_{\mathrm{T}} = |\vec{p}| \sin\theta$ is the momentum component perpendicular to the beam. Before the collision the transverse momentum is zero, so p_{T} is the natural scale for the hard products of a collision. The eFEX estimates $p_{\mathrm{T}}^{\mathrm{TOB}}$ from the calorimeter energy of the object (in GeV, in steps of $100\,\mathrm{MeV}$, with a minimum of $5\,\mathrm{GeV}$). The true p_{T} of a τ comes from the simulation and is not available in real data.
- **Angular distance** between two objects is $\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2}$, with $\Delta\phi$ wrapped to $[-\pi, \pi)$.

In the CP inputs, η and ϕ are taken relative to the highest-$p_{\mathrm{T}}^{\mathrm{TOB}}$ object of the event, so they describe where an object lies with respect to the leading activity. TOB positions are given at tower resolution. The BDT score is a per-TOB boosted-decision-tree output supplied with the eFEX objects.

A.6 Per-object columns

Table 4 lists every column available per object and how it may be used. Energies are stored in MeV and converted to GeV for evaluation. The 12×12 EM2 image sums exactly to the EM2 3×3 core, so the two are views of the same energy at different granularity. For truth-matched τ objects the median of $p_{\mathrm{T}}^{\mathrm{TOB}}/p_{\mathrm{T}}^{\mathrm{true}}$ is 0.59, which reflects the energy carried away by the neutrino and the TOB energy scale.

Table 4: Columns per object and how they may be used. "Input" columns may feed a network. "Truth" columns are used only for labels, loss weights and offline evaluation.

Column	Content	Use
`X_tensors`	core tensor, $5 \times 3 \times 3$ cell energies (EM0–EM3, HAD)	input
`X_em2_tensors`	EM2 image, 12×12 cell energies	input
`tob_pt`	eFEX TOB transverse momentum ($100\,\mathrm{MeV}$ steps, $\geq 5\,\mathrm{GeV}$)	input / baseline
`tob_eta`, `tob_phi`	TOB position (tower resolution); used relative to the leading TOB	input
`tob_bdt`	eFEX BDT score	input
`eventNumber`, `tob_index`	event identifier and TOB slot (0–5)	grouping only
`Type`	sample of origin (signal or background)	grouping only
`signal`	truth match to a hadronic τ	truth (label)
`truth_pt`	true τ transverse momentum	truth (weights, eval.)
`prongs`	number of charged decay products	truth (analysis)

A.7 Event-level splits

The counts below refer to the retained analysis dataset rather than the total generated production. Events are identified by (sample, event number), since identifiers can repeat between the signal and background samples. Each of the three seeds shuffles the complete events and splits them $70/10/20\%$ into training, validation and test sets. Table 5 shows that every split reproduces the composition of the full dataset: the fraction of signal events stays within 0.6 percentage points and the median true τ p_{T} within $0.5\,\mathrm{GeV}$ of the full-sample values.

A.8 Dataset statistics

The median true p_{T} of truth τ objects is $41.9\,\mathrm{GeV}$. The quartiles are 25.6 and $84.1\,\mathrm{GeV}$. Above $25\,\mathrm{GeV}$ the spectrum falls approximately as t^{-2}: in the training split the 25–30 GeV bin contains 8.3 times more τ objects than the 95–

Table 5: Composition of every event split. Signal-event fraction, truth-τ fraction among all objects, and median true p_{T} of truth τ objects.

	Signal events [%]			τ / objects [%]			Median τ p_{T} [GeV]		
Seed	train	val	test	train	val	test	train	val	test
42	22.48	22.73	22.57	6.12	6.19	6.15	42.0	42.2	41.4
123	22.50	23.06	22.35	6.13	6.26	6.09	41.9	42.0	42.0
456	22.52	22.41	22.60	6.13	6.11	6.15	41.9	42.1	41.7
Full dataset		22.53			6.13			41.9	

Table 6: Event-level split for seed 42. Complete events are assigned to one split. The last column is the largest number of background events that may fire under the 0.5% event-level budget.

Split	Signal events	Background events	Objects	Truth τ objects	Max. firing bkg. events
Train	50,588	174,410	1,265,650	77,498	872
Validation	7,307	24,835	180,757	11,184	124
Test	14,509	49,777	361,685	22,239	248
Total	72,404	249,022	1,808,092	110,921	–

100 GeV bin. The test set contains 5,145, 5,341, 3,705, 3,199 and 794 truth τ objects in the bands 10–25, 25–40, 40–60, 60–100 and 100–120 GeV. The validation set contains 2,458, 2,714, 1,937, 1,548 and 391. Above 60 GeV a 5 GeV validation window holds only 90–250 truth τ objects, so a single object moves its efficiency by 0.4–1.1 percentage points. This resolution limit shaped several design choices (Appendices D and F).

A.9 Energy-conditioned compactness

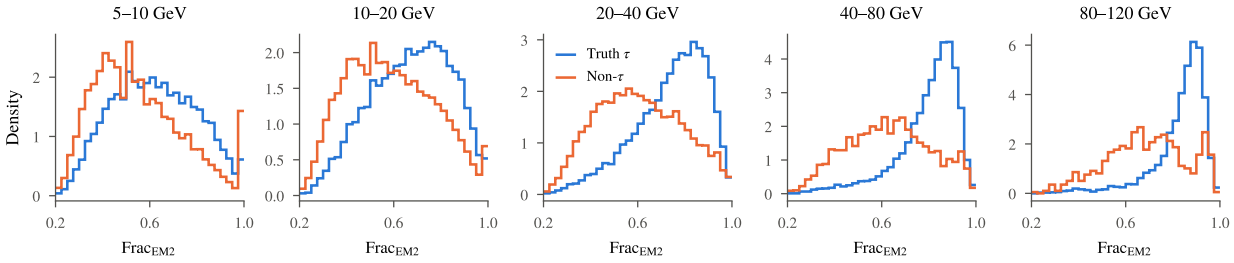


Figure 6: $\mathrm{Frac}_{\mathrm{EM2}}$ for truth τ and non-τ objects in fixed $p_{\mathrm{T}}^{\mathrm{TOB}}$ intervals (training split, normalised per class). The separation remains inside every interval and grows with energy, so the feature carries shape information beyond the energy scale.

B Feature Definitions

Let e be an event and i an object in it. For layer $\ell \in \{\mathrm{EM0}, \mathrm{EM1}, \mathrm{EM2}, \mathrm{EM3}, \mathrm{HAD}\}$, $E_{\ell,ab}$ is the energy of tower (a, b) of its 3×3 window. E_{ab}^{12}, $a, b \in \{0, \ldots, 11\}$, is the EM2 image. When a ratio has an exactly zero denominator, the implementation returns zero. Every entry below is a key of the feature registry and is selected through the experiment configuration.

B.1 Input bundles

- `core_tensors` (CT): $x_i = \mathrm{vec}(E_{\mathrm{EM0}}, E_{\mathrm{EM1}}, E_{\mathrm{EM2}}, E_{\mathrm{EM3}}, E_{\mathrm{HAD}}) \in \mathbb{R}^{45}$. Cell $9L + 3r + c$ is layer L, row r, column c.
- `core_physics` (CP): $x_i = [\mathrm{BDT}_i,\ p_{\mathrm{T},i}^{\mathrm{TOB}},\ \eta_i - \eta_{\mathrm{lead}},\ \mathrm{wrap}(\phi_i - \phi_{\mathrm{lead}})]$, where "lead" is the highest-$p_{\mathrm{T}}^{\mathrm{TOB}}$ object of the event and wrap maps an angle to $[-\pi, \pi)$.
- `em2_all_cells` (EM2): $\mathrm{vec}(E^{12}) \in \mathbb{R}^{144}$, reshaped to $1 \times 12 \times 12$ by the convolutional models.
- `tob_pt_only`: $x_i = p_{\mathrm{T},i}^{\mathrm{TOB}}$.

B.2 Per-layer summaries

For every layer, with $S_\ell = \sum_{a,b} E_{\ell,ab}$ and $M_\ell = \max_{a,b} E_{\ell,ab}$,

$$D_\ell = M_\ell - (S_\ell - M_\ell) = 2M_\ell - S_\ell, \qquad Q_\ell = \sum_{a,b} \mathbf{1}\big[E_{\ell,ab} > 0.1\, M_\ell\big]. \tag{11}$$

S_ℓ is the energy sum, D_ℓ the core dominance (the maximum cell minus its neighbours) and Q_ℓ the dynamic sparsity (the number of active cells). The 15 values $\{S_\ell, D_\ell, Q_\ell\}_\ell$ form the SUM bundle. These definitions avoid cell-by-cell division, which was not available in the firmware specification of [5].

B.3 Full-granularity EM2 features

With $S^{12} = \sum_{a,b} E^{12}_{ab}$, $(a^\star, b^\star) = \arg\max_{a,b} E^{12}_{ab}$ and $d^2_{ab} = (a - a^\star)^2 + (b - b^\star)^2$:

$$\mathrm{Frac}_{\mathrm{EM2}} = \frac{\max_{w \in \mathcal{W}_{3\times 3}} \sum_{(a,b)\in w} E^{12}_{ab}}{S^{12}}, \qquad W_{\mathrm{norm}} = \frac{\sum_{a,b} E^{12}_{ab}\, d^2_{ab}}{S^{12}}, \tag{12}$$

the compactness (`em2_best_3x3_fraction`) and the normalised shower width (`em2_normalized_width`). Further entries include the largest and second-largest cells, the squared distance between them (`em2_maxdist`), the energy outside the best window divided by $p_{\mathrm{T}}^{\mathrm{TOB}}$, and the energies and centre distances of the three strongest non-overlapping 3×3 windows. The registry also contains versions computed on the 3×3 EM2 core, and approximations that use subtraction instead of division. `em2_3x3_maxdist` is highly discrete. An early result that ranked it first is excluded: the earlier evaluation suffered from discrete-threshold defects, so that result was not a valid 0.5% comparison. All results in this paper use the tie-aware calibration of Appendix C.

B.4 Event-level and object-geometry features

With $r_1(e), r_2(e)$ the two leading objects:

$$H_T^{\mathrm{TOB}} = \sum_{i \in e} p_{\mathrm{T},i}^{\mathrm{TOB}}, \qquad p_{\mathrm{T},r_2(e)}^{\mathrm{TOB}}, \qquad \Delta R_{12} = \sqrt{(\eta_{r_1} - \eta_{r_2})^2 + \mathrm{wrap}(\phi_{r_1} - \phi_{r_2})^2}. \tag{13}$$

These are computed once per event and copied to every object. For object-specific geometry, the partner $p(i)$ of object i is the leading object other than i. The feature `object_reference_tob_dr2` is $(\eta_i - \eta_{p(i)})^2 + \mathrm{wrap}(\phi_i - \phi_{p(i)})^2$. The feature `object_partner_context` is $[\log(1 + p_{\mathrm{T},i}^{\mathrm{TOB}}), \log(1 + p_{\mathrm{T},p(i)}^{\mathrm{TOB}}), \pi - |\Delta\phi_{i,p}|, |\Delta\eta_{i,p}|]$, which keeps the partner energy, the azimuthal acoplanarity and the η separation separate. In the distribution study, events with two or more τ objects have a much harder second-highest $p_{\mathrm{T}}^{\mathrm{TOB}}$ and a larger H_T^{TOB} than events without a τ. Their squared separation ΔR_{12}^2 also peaks near π^2, the back-to-back topology. Background events populate the same region too, however. In our single-object tests, the event sum and the object-geometry features did not improve efficiency at ε_{B} over object-level inputs.

C Event-Level Calibration and Statistical Tools

C.1 Metrics, baseline and reporting protocol

The signal efficiency $\epsilon(p_{\mathrm{T}})$ is the fraction of truth-matched tau objects accepted at the calibrated threshold, rather than the fraction of signal events firing the trigger. It is reported globally over all test tau objects and in the true-p_{T} bands 10–25, 25–40, 40–60, 60–100 and 100–120 GeV. The event-level fake rate instead counts firing background events. The baseline accepts an object when $p_{\mathrm{T}}^{\mathrm{TOB}} \geq C$ and uses the same two-object event decision as the network. Although this is a hard cut in reconstructed $p_{\mathrm{T}}^{\mathrm{TOB}}$, its efficiency versus true p_{T} turns on gradually because the reconstructed response is shifted and smeared; some high-true-p_{T} objects still fall below the cut.

For the reported fixed-rate comparisons, the model and baseline thresholds are recalibrated on the same test background at the same target, and both achieved rates are reported. This measures discrimination at a common operating point; it does not test transfer of a frozen threshold. Performance using the validation-calibrated threshold is retained separately for that purpose. Test recalibration does not select features, losses, budgets, architectures or checkpoints. The design goal is non-inferiority above 25 GeV, earlier saturation and gains across 25–100 GeV, with some loss below 25 GeV acceptable. These are goals, not a claim that every reported model satisfies them. Accuracy, ROC-AUC and minimum BCE are not the primary selection criteria.

C.2 Tie-aware threshold selection

Let $\mathcal{B}$ be a set of N background events. For the two-object trigger, the *event trigger score* is $u_e = s_{e,(2)}$, the second-highest object score. An event with fewer than two objects can never fire. The procedure is:

1. Let $v_1 > v_2 > \cdots > v_m$ be the distinct values of $\{u_e\}$, and let $c_j = |\{e : u_e \geq v_j\}|$ be the cumulative counts (all events tied at a value are counted together).
2. Let $k_{\max} = \lfloor \varepsilon_{\mathrm{B}} N \rfloor$. Return $t^\star = v_{j^\star}$ with $j^\star = \max\{j : c_j \leq k_{\max}\}$. If no such j exists, return the next floating-point value above v_1, so that no event fires.
3. Report the *achieved* rate $\widehat{\mathrm{FPR}} = |\{e : u_e \geq t^\star\}|/N \leq \varepsilon_{\mathrm{B}}$.

Because the search runs over distinct values, a group of tied scores is never split: all of its events pass or all fail. A threshold between two scores could claim a fake rate the deployed comparison $s \geq t$ cannot reproduce. Calibration and deployment use the same float64 comparison.

C.3 Joint calibration of the OR classifier

For a TOB budget b, the TOB cut C_b is obtained by applying the procedure above to the second-highest $p_{\mathrm{T}}^{\mathrm{TOB}}$ with target b. Given C_b, let $n_e = \sum_i \mathbf{1}[q_{e,i} \geq C_b]$ be the number of objects the TOB branch accepts in event e.

- Events with $n_e \geq 2$ fire regardless of the network. They form a fixed base count N_0.
- In an event with $n_e < 2$, the network must accept $2 - n_e$ of the remaining objects. The *activation score* u_e^{OR} is the $(2 - n_e)$-th highest network score among the objects the TOB branch did not accept. The event fires if and only if $u_e^{\mathrm{OR}} \geq t$.

The network threshold t_b is the tie-aware threshold for the scores $\{u_e^{\mathrm{OR}}\}$ with the budget reduced by the base count, $k_{\max} - N_0$. This construction counts mixed events, in which one object passes each branch, exactly.

C.4 Cross-fitting

A threshold calibrated and evaluated on the same background events reports a fake rate of almost exactly ε_{B} by construction, and hides how well the threshold transfers to new events. Selection steps that compare many calibrated candidates therefore use two-fold cross-fitting. The events (never single objects) are split into two deterministic folds, stratified by class. The threshold is calibrated on fold A and all rates are measured on fold B, then the roles are swapped and the counts are pooled. This applies to the choice of the TOB budget, to checkpoint selection in constrained training, and to the dual updates on the training constraint subset. The plain target-FPR checkpoint rule (Appendix H) calibrates and measures on the full validation set. After selection, one deployment threshold is calibrated on the complete validation set.

C.5 Event-clustered paired uncertainty

Guards compare a candidate classifier with a reference (the baseline or the pretrained model) on the *same* truth τ objects of a region r. Let $d_i = \mathbf{1}[\text{candidate accepts } i] - \mathbf{1}[\text{reference accepts } i]$, let n be the number of objects, $\hat{\Delta}_r = \frac{1}{n}\sum_i d_i$, and $D_e = \sum_{i \in e} d_i$ with n_e objects in event e. Objects of one event are correlated, so the standard error is clustered by event (G events):

$$\hat{\sigma}_r^2 = \frac{G}{G-1} \frac{\sum_e \left(D_e - \hat{\Delta}_r n_e\right)^2}{n^2}, \qquad L_r = \hat{\Delta}_r - z\,\hat{\sigma}_r. \tag{14}$$

L_r is a one-sided lower confidence bound. The OR budget search uses $z = 1$, and constrained-training selection uses $z = z_{0.95}$.

C.6 Confidence-safe deployment target

Let $U(k, N)$ be the exact one-sided Clopper–Pearson upper bound [16] on a binomial rate with k firing events out of N, $U(k, N) = \mathrm{Beta}^{-1}(1-\alpha;\, k+1,\, N-k)$. The confidence-safe target is $k^\star/N$, where $k^\star$ is the largest $k \leq \lfloor \varepsilon_{\mathrm{B}} N \rfloor$ with $U(k, N) \leq \varepsilon_{\mathrm{B}}$. The deployed threshold is calibrated to this lower empirical rate. With the validation statistics available, the one-sided 95% target corresponds to an empirical fake rate of about 0.42%.

D The TOB–NN OR Classifier

D.1 From a mixture of experts to a logical OR

With plain BCE and $p_{\mathrm{T}}^{\mathrm{TOB}}$ among the inputs, the learned turn-on curve often resembled the TOB baseline from below. The network learned most of the energy dependence but not enough shape information to beat the baseline at high energy. Our first remedy was a two-expert model: an energy expert with inputs such as $p_{\mathrm{T}}^{\mathrm{TOB}}$, and a shape expert with $\mathrm{Frac}_{\mathrm{EM2}}$, widths and angular variables. The two were combined as $s_{\mathrm{mix}} = \alpha s_{\mathrm{energy}} + (1-\alpha) s_{\mathrm{shape}}$, with α and the threshold chosen on validation. Preliminary runs gave no meaningful gain, and the energy expert only relearned a rule that the eFEX already implements exactly. The OR classifier (4) keeps this rule deterministic and gives the network the complementary task.

D.2 Joint calibration

Given a TOB budget b, the comparator threshold C_b is set so that the TOB branch alone fires in at most a fraction b of background events. The NN threshold t_b is then calibrated so that the complete OR fires in at most ε_{B}. Both steps operate on event decisions: an event containing one TOB-accepted object and one distinct NN-accepted object must count as accepted, even when neither branch alone would fire. The budget is selected separately for each model on validation, not assigned during training.

D.3 Per-epoch selection procedure

At the end of every training epoch:

1. the network is trained with (weighted) BCE; the OR rule does not enter the gradient;
2. validation scores are predicted;
3. for every candidate $b \in \mathcal{B} = \{0, 0.0005, \ldots, 0.005\}$, the pair (C_b, t_b) is calibrated with two-fold event cross-fitting inside the validation set (Appendix C);
4. candidates that exceed the 0.5% event fake rate or violate a protection guard are discarded;
5. the feasible candidate with the largest objective $J(b)$ (16) is selected; when two objectives differ by at most 0.002, the one with the safer worst-window margin is preferred;
6. epochs are compared with the same objective.

The minimum-BCE checkpoint is kept as a reference. No budget, threshold or epoch is chosen on test data. The procedure is repeated independently for every feature set, seed and checkpoint, because the score distribution of the network and its overlap with the TOB branch change with the inputs.

D.4 Objective

The target is the mean efficiency advantage over the region of interest,

$$J_{\mathrm{cont}}(b) = \frac{1}{100-25} \int_{25}^{100} \left[\epsilon_{\mathrm{OR}}(p; b) - \epsilon^{\mathrm{TOB}}(p)\right] dp. \tag{15}$$

It is estimated over $5\,\mathrm{GeV}$ windows spanning 25–100 GeV:

$$J(b) = |\mathcal{K}|^{-1} \sum_{k \in \mathcal{K}} \left[\hat{\epsilon}_{\mathrm{OR},k}(b) - \hat{\epsilon}_k^{\mathrm{TOB}}\right]. \tag{16}$$

Equal window weights state the physics preference directly: a gain in a sparsely populated high-energy window counts as much as one near the threshold.

D.5 Protection guards

Per window. The original rule required $\hat{\Delta}_k \geq -0.005$ in every $5\,\mathrm{GeV}$ window up to $120\,\mathrm{GeV}$. It proved too sensitive to fluctuations: saturated validation windows contain only 90–250 truth τ objects, so one object moves the efficiency by 0.4–1.1 points, and a candidate could be accepted or rejected because of a single object.

Pooled saturation. The $5\,\mathrm{GeV}$ guards are kept below $60\,\mathrm{GeV}$, and all guards above are replaced by one pooled region with $\hat{\Delta}_{60:120} \geq -0.005$. Pooling increases the sample, but a local deficit can be averaged away.

Multiscale saturation. The fine guards are kept below $60\,\mathrm{GeV}$. The saturation range is tested on the overlapping regions $[60, 90)$, $[70, 100)$, $[80, 110)$, $[90, 120)$ and the full $[60, 120)$, each with the event-clustered lower bound $L_r = \hat{\Delta}_r - z\hat{\sigma}_r \geq -\delta_{\mathrm{phys}}$ ($z = 1$, $\delta_{\mathrm{phys}} = 0.005$). The overlapping regions detect a persistent local loss that pooling would hide, and they are more stable than $5\,\mathrm{GeV}$ windows. A tolerance of $\delta_{\mathrm{phys}} = 0$ was needlessly strict, because it demanded a statistically positive gain even where both classifiers are already close to full efficiency.

D.6 Interpreting the selected budget

A small $b^\star$ leaves more of the background budget to decisions introduced by the network, and usually indicates a larger independent contribution from the NN. A large $b^\star$ indicates stronger reliance on the comparator. This reading is useful but not absolute, because $b^\star$ also depends on score calibration, seed fluctuations and the overlap of the two branches. The budget changes the final decision rule, whereas signal weighting (Appendix E) changes the learned score. The two are separate mechanisms and were studied separately.

E Energy-Weighted BCE

For an object with label y_i, score s_i and weight w_i,

$$\mathcal{L}_w = -\frac{1}{N}\sum_{i=1}^{N} w_i \big[y_i \log s_i + (1 - y_i)\log(1 - s_i)\big]. \tag{17}$$

Background objects always keep $w_i = 1$. They have no meaningful true τ momentum, and reweighting them would distort the rare background tail that sets the 0.5% event fake rate. Signal weights are built from a raw profile $r(t)$ of the true momentum t (in GeV) and normalised on the *training* signal only:

$$w_i = \frac{r(t_i)}{\frac{1}{N_S}\sum_{j\in\mathcal{S}_{\mathrm{train}}} r(t_j)}, \qquad y_i = 1. \tag{18}$$

The mean training signal weight is therefore one, so a profile redistributes the signal gradient across energy without changing the overall signal–background balance. A profile fitted on training data is reused unchanged on validation data. Truth momentum is used only to build weights, never as an input.

E.1 Turn-on emphasis (α profile)

The first hypothesis was that the TOB branch already protects the plateau, so the network should concentrate on the turn-on:

$$r_\alpha(t) = \begin{cases} 1, & t < 25, \\ 1 + \alpha, & 25 \leq t < 40, \\ 1 + \alpha/2, & 40 \leq t < 80, \\ 1, & t \geq 80, \end{cases} \qquad \alpha \in \{0, 1, 2, 4\}. \tag{19}$$

$\alpha = 0$ is exactly plain BCE and serves as a control. Every α is a separate training run with α fixed throughout.

E.2 Inverse-frequency profile

The range 25–$100\,\mathrm{GeV}$ is divided into $5\,\mathrm{GeV}$ bins. With n_k training-signal objects in populated bin $k \in \mathcal{K}_+$,

$$\bar{n} = \frac{1}{|\mathcal{K}_+|}\sum_{k\in\mathcal{K}_+} n_k, \qquad r_k = \mathrm{clip}\left(\frac{\bar{n}}{n_k},\, 0.2,\, 5\right), \tag{20}$$

and $r = 1$ outside the range. Without clipping, $n_k r_k \approx \bar{n}$: every bin contributes the same total signal weight. The clipping keeps a nearly empty bin from producing an unstable gradient. For seed 42 the fitted weights are about 0.39 at 25–30 GeV, 0.57 at 35–40, 0.96 at 50–55, 1.36 at 60–65, 2.06 at 75–80 and 3.28 at 95–100 GeV.

E.3 Power-law profile

Above $\approx 25\,\mathrm{GeV}$ the training τ spectrum is well described by $f_S(t) \approx c/t^2$. The unweighted signal contribution of a narrow interval $[t_0, t_0 + dt]$ to BCE is then

$$d\mathcal{L}_S(t_0) \propto \frac{1}{t_0^2}\,\mathbb{E}\big[\ell_{\mathrm{BCE}} \mid t_0\big]\, dt. \tag{21}$$

A 25–30 GeV bin contains about eight times as many training taus as a 95–100 GeV bin. Even when the average loss per object is similar at all energies, soft τ objects dominate the gradient because there are many more of them. Plain BCE therefore already favours the dense, low-energy part of the spectrum. For a one-parameter control over this effect, we clamp $\tilde{t}_i = \min(200, \max(10, t_i))$ and set $r_p(t_i) = \tilde{t}_i^{-p}$. Away from the clamp,

$$f_S(t)\, w_p(t) \propto t^{-(2+p)}. \tag{22}$$

- $p = 0$: plain BCE.
- $p > 0$: relatively more weight on low-p_{T} τ objects.
- $p < 0$: relatively more weight on high-p_{T} τ objects.
- $p = -1$: equal weighted signal counts per logarithmic interval, since $t^{-1}dt = d\log t$.
- Inverse frequency: for equal-width bins and $f_S \propto t^{-2}$, it behaves roughly like a clipped t^2 weight, i.e. qualitatively $p \approx -2$.

IF is not the power-law profile at $p = -1$ (or exactly at $p = -2$): it uses empirical counts, different clipping and unit raw weights outside 25–100 GeV. The clamp keeps very small or very large momenta from producing extreme weights. The screen covered $p \in \{-1.9, -1.6, -1.3, -1, -0.6, -0.3, 0.3, 0.6, 1\}$, together with plain BCE and inverse frequency. The negative side was extended after the first screen favoured more weight on high-energy τ objects.

E.4 Why emphasising energetic taus can help soft ones

We propose a mechanism, not a proof. Energetic τ showers have a higher signal-to-noise ratio, more stable shape measurements and less overlap with background, so they are cleaner examples of the class. A representation learned from them can transfer to weaker, low-energy showers. Strongly up-weighting soft objects instead emphasises noisy or intrinsically ambiguous examples, raises the gradient variance and moves the decision boundary without teaching a more general τ signature. The mechanism is consistent with the smaller TOB budgets selected under inverse-frequency weighting (Appendix D). No weight profile is optimal in general: the best profile depends on the features, the model capacity, the optimisation and the class overlap.

F The Constrained Trigger Objective

F.1 Problem

Let $s_i = f_\theta(x_i)$. Under the NN-only classifier an object passes when $s_i \geq t$. Under the OR classifier it passes when $(p_{\mathrm{T},i}^{\mathrm{TOB}} \geq C) \vee (s_i \geq t)$. The same event rule (1) applies in both cases. One list of energy regions should not serve two purposes: improvement is wanted where there is room for it, while protection is needed everywhere. The objective regions and weights are

$$\mathcal{K}_J = \{[25, 32), [32, 40), [40, 60)\}\,\mathrm{GeV}, \qquad w = (0.35, 0.35, 0.30), \quad \textstyle\sum_k w_k = 1, \tag{23}$$

and the constraint regions are $\mathcal{K}_C = \mathcal{K}_J \cup \{[60, 120)\}$. The turn-on window is split at 32 GeV because the measured advantage changes strongly inside $[25, 40)$. The saturated region is left out of the objective: its attainable gain is bounded by $1 - \epsilon_{60:120}^{\mathrm{TOB}} \approx 0.03$, and its measured advantage is comparable to its statistical uncertainty. The weights encode a fixed scientific preference and are not learned, because an optimiser allowed to choose them would put all weight on the easiest region. Each constraint region carries the floor

$$r_k = \min\Big[1,\ \max\big(\epsilon_k^{\mathrm{TOB}} + m_k,\ \epsilon_k^{(0)} - \rho_k\big)\Big], \tag{24}$$

where m_k is the required advantage over the baseline, $\epsilon_k^{(0)}$ is the efficiency of the frozen pretrained network, and ρ_k is the largest loss allowed relative to it. We use $m = (0.0025, 0.0025, 0.0025, 0)$ and $\rho = (0.0025, 0.0025, 0.0025, 0.005)$. The second term is needed because fine-tuning starts from a strong model: a guard that protects only the baseline allowed the network to give up part of an advantage it already had. The complete problem is

$$\max_\theta\ J(\theta) = \sum_{k \in \mathcal{K}_J} w_k \big[\epsilon_k(\theta) - \epsilon_k^{\mathrm{TOB}}\big] \quad \text{s.t.} \quad \mathrm{FPR}_{\mathrm{ev}}(\theta) \leq 0.005, \quad \epsilon_k(\theta) \geq r_k \ \ \forall k \in \mathcal{K}_C. \tag{25}$$

F.2 Differentiable surrogate

The hard threshold and the "at least two" count have zero gradient almost everywhere. The network update therefore uses smooth surrogates, while every constraint measurement uses the exact hard decisions. The construction replaces

an object step by a temperature-controlled sigmoid and the event count by the probability of at least two successes under the surrogate's independent Bernoulli decisions. This is a differentiable device, not an assumption that physical objects within an event are independent. As $T \to 0$, it approaches the hard decision away from threshold ties.

Soft object decision. $q_i = \sigma\big((z_i - t_{\text{proxy}})/T\big)$, where z_i is the pre-sigmoid score (logit) of object i. For the OR classifier the comparator stays hard: $q_i^{\text{OR}} = h_i + (1 - h_i)q_i$ with $h_i = \mathbf{1}[p_{\text{T},i}^{\text{TOB}} \geq C]$.

Rank-calibrated proxy threshold. In every batch, t_{proxy} is recomputed as the empirical background quantile that matches the 0.5% event target, taken from the ranked second-highest object logits of the background events in the batch. A common shift of all scores then moves the threshold by the same amount, so the score-inflation direction carries no gradient.

Temperature annealing. The gradient of q_i is proportional to $q_i(1-q_i)/T$ and becomes negligible for $|z_i - t_{\text{proxy}}| \gtrsim 3T$. A small temperature T gives a faithful surrogate that sees only a narrow band around the boundary. A large T reaches distant objects but optimises a blurred trigger. T therefore follows a cosine schedule from 0.10 to 0.02.

Event surrogate. The probability that at least two objects pass is the Poisson-binomial tail

$$\tilde{P}_e^{(2)} = 1 - \prod_{i \in e}(1 - q_i) - \sum_{i \in e} q_i \prod_{j \in e,\, j \neq i}(1 - q_j). \tag{26}$$

With $\mathcal{B}$ the background events and $\mathcal{S}_k$ the truth τ objects of region k, $\widetilde{\text{FPR}} = |\mathcal{B}|^{-1} \sum_{e \in \mathcal{B}} \tilde{P}_e^{(2)}$ and $\tilde{\epsilon}_k = |\mathcal{S}_k|^{-1} \sum_{i \in \mathcal{S}_k} q_i$. These quantities are used only for optimisation and never to certify a constraint.

Why recalibration matters. With a frozen proxy threshold, the optimiser could increase every score through the output bias, improving the soft objective without improving the deployed ranking; the fake rate at that frozen threshold grew several-fold. Recomputing the proxy threshold from background ranks makes it move with a common score shift and removes this artificial gain. This addresses score translation; it does not imply that a fixed-temperature sigmoid surrogate is invariant to every monotone score transformation. Annealing then trades broad early gradient coverage for a sharper late approximation.

F.3 Primal–dual training

The Lagrangian is

$$\mathcal{L}(\theta, \lambda) = -\tilde{J}(\theta) + \lambda_{\text{FPR}} \frac{\widetilde{\text{FPR}}(\theta) - 0.005}{0.005} + \sum_{k \in \mathcal{K}_C} \lambda_k \big(r_k - \tilde{\epsilon}_k(\theta)\big), \tag{27}$$

where $\tilde{J}$ replaces each hard efficiency by its surrogate. The fake-rate violation is expressed relative to its target so that the fake-rate and region terms have comparable scales. Without this normalisation, a one-per-mille excess in fake rate would be numerically invisible next to an efficiency deficit of the same physical importance. The two updates alternate. The network parameters take Adam steps on $\mathcal{L}$. The multipliers follow projected dual ascent on *hard* violations g_j,

$$\lambda_j \leftarrow \big[\lambda_j + \eta_j\, g_j\big]_{[0, \lambda_{\max}]}, \qquad \lambda_{\max} = 25, \quad \eta_{\text{FPR}} = 1, \quad \eta_k = 50. \tag{28}$$

All multipliers start at zero and become active only after a measured violation. The region rates are larger because region violations are numerically small: they correspond to a handful of signal objects. The hard violations are measured once per epoch on a held-out *constraint subset* of the training events, with two-fold cross-fitting. This keeps optimistic in-sample calibration from driving the multipliers: in earlier stages the in-sample measurement reported comfortable margins while the validation margins were already negative, and every λ_k stayed at zero throughout training. The division of labour is strict: differentiable quantities provide gradients, and exact deployable decisions determine whether a constraint is satisfied.

F.4 Protocol and leakage protection

Constrained training is a fine-tuning stage. It starts from the best weighted-BCE checkpoint of the same input representation and runs for 20 epochs at learning rate 10^{-4}, with batches of 2048 complete events. The training events are split into two disjoint, stratified subsets: a primal subset (70%) for the network update and a constraint subset (30%) used only for the dual update. Complete events stay together, because the event fake rate cannot be estimated from shuffled objects. Checkpoint selection uses validation cross-fitting (Appendix C). The held-out validation fake rate measured this way is typically 0.35–0.40%, while the in-sample value is 0.499%. After the best epoch is chosen, one deployment threshold is calibrated on the full validation set with the confidence-safe target. The test set is never used for training, calibration or selection.

F.5 Uncertainty-aware feasibility

Feasibility based on point estimates was fragile in both directions. A checkpoint could be accepted because a window of 90 objects fluctuated upward, or rejected because it fluctuated downward. Every constraint is therefore evaluated with its event-clustered lower bound L_r (Appendix C). The first fully certified rule required $L_r \geq m_k$ in every region and a certified fake-rate bound below target. It was too strict: certifying a 0.5% rate at 95% confidence requires operating near 0.42%, and certifying a non-negative advantage in the saturated region requires an observed advantage of about $+0.8$ points where the true value is compatible with zero. Even the initial model was infeasible, and two of three seeds returned their initialisation. The corrected rule keeps the uncertainty machinery but (i) applies the fake-rate test to the measured cross-fitted rate and records the Clopper–Pearson bound as a diagnostic (the budget itself is protected by the confidence-safe deployment target), and (ii) includes the tolerance in the certified guards, $L_r \geq m_k - \delta_{\mathrm{phys}}$ with $\delta_{\mathrm{phys}} = 0.005$, so a saturated region must be statistically compatible with non-inferiority rather than certifiably superior. Among feasible epochs, the one with the largest cross-fitted hard objective is selected. If no epoch is feasible, the one with the smallest certified violation is kept.

F.6 Applying the loss to other inputs and to the CNNs

The objective, the constraints, the surrogates and the dual updates act only on network scores and measured efficiencies, and no term refers to the input variables. A different input representation is therefore a configuration change, not a code change. Three surrounding components are matched to the representation and are prepared for each one:

- **Initialisation and reference guard.** Fine-tuning starts from the best weighted-BCE model of the *same* inputs: inverse frequency for CT and SUM+F, and $p = -1$ for CP+CT and CT+F. The reference floor (24) is defined relative to this starting point, so a checkpoint trained on other inputs cannot be reused.
- **Temperature scale.** T acts on the score scale, and different inputs produce score distributions of different width near the threshold. The fraction of objects inside the effective gradient band $|z_i - t_{\mathrm{proxy}}| \lesssim 3T$ is recorded per region before training. The annealing endpoints are changed only if this coverage differs strongly from the reference.
- **What transfers unchanged.** Across the MLP inputs, the regions, weights and tolerances (m_k, ρ_k, δ_{phys}) are physics choices in efficiency units. The dual rates and the fake-rate normalisation act on rate scales. The cross-fitting, the confidence-safe calibration and the selection protocol are independent of the inputs.

The same recipe is used for the convolutional models, where "FT" denotes a constrained fine-tune of an IF or BCE model (Table 7). Their weak end was saturation rather than the turn-on, so for them the objective also rewards $[60, 100)$ GeV, with weights $w = (0.30, 0.30, 0.25, 0.15)$, and the saturated region is guarded as two regions, $[60, 100)$ and $[100, 120)$ GeV, each allowed at most 0.5 points below the starting model. This is the only change of regions or weights between input representations.

F.7 Discarded variants

Frozen proxy threshold with a counterweight. Keeping t_{proxy} fixed and relying on λ_{FPR} (initialised by balancing gradient norms) to suppress score inflation only moved the equilibrium: each seed settled at a different inflation level. The rank-calibrated threshold removes the direction from the loss itself.

Tail-ranking objective. A pairwise logistic loss compared truth-τ logits with the second-highest logits of the top 5% of background events, selected by rank. It is shift invariant and gives full gradient to signal far below the boundary. However, it raised the low-energy gain faster than the saturation constraints could respond, and checkpoint selection fell back to early epochs.

Hard-negative memory bank. A buffer of median-centred background offsets from earlier batches was meant to enlarge the ranking tail. The stale offsets no longer matched the current network, and all seeds selected their initialisation.

F.8 Requirements for a trigger loss

These requirements emerged from failed variants as well as successful fine-tuning. Weighted BCE captured much of the available improvement; the custom objective chiefly made energy-dependent trade-offs explicit. The lessons concern the complete loss, calibration and selection procedure, not a claim that one surrogate is universally optimal.

(1) **Respect the calibrated ranking.** After threshold recalibration, a common strictly increasing transformation of the scores preserves their ranking and hence the hard decision at a fixed rate, apart from numerical or tie effects. A surrogate should not reward changes that merely move all scores together. With a frozen proxy threshold, output-bias inflation increased the training fake rate from 0.5% to above 3% without improving separation. A minimal

diagnostic is to add a constant to every logit and verify that the objective does not improve. Recomputing the background-quantile threshold removes this translation shortcut; it does not make a fixed-temperature sigmoid invariant to arbitrary monotone transformations.

(2) **Make regional trade-offs explicit.** Equal object weights are not equal energy-region weights: the falling spectrum gives abundant soft taus more influence. A region absent from both reward and protection can pay for gains elsewhere. In the SUM+F fine-tune, improvements at 32–60 GeV accompanied a 0.69-point loss at 10–25 GeV, which had neither role. Every scientifically relevant region therefore needs an explicit reward, floor or deliberately accepted trade-off. Adaptive constraint multipliers cannot protect a region omitted from the formulation.

(3) **Inspect gradient coverage, not just the loss value.** A narrow sigmoid surrogate can be accurate near the threshold yet give almost no gradient to missed taus far below it. In one examined setting, their gradient magnitude was about $1/4600$ of that at the boundary. Conversely, the tail-ranking variant reached distant signal but changed the efficiency profile faster than the protection mechanism could react. Inspect gradient magnitude against score and energy, including missed signal; temperature annealing balances early reach against late decision fidelity.

(4) **Optimise softly, assess hard decisions out of sample.** Differentiability is a training device, not evidence of constraint satisfaction. Threshold calibration and violation measurement must use separate event subsets through cross-fitting, and measurements must use the exact two-object trigger. Earlier in-sample checks reported safe regional margins while held-out margins were negative, leaving all regional multipliers inactive. Held-out constraint events restored meaningful feedback; validation remained separate for checkpoint selection.

(5) **Match tolerances to statistical resolution.** Sparse windows cannot support arbitrarily fine protection thresholds: one accepted tau among 90–250 objects changes efficiency by roughly 0.4–1.1 percentage points. The first fully certified rule rejected even the initial model; pooled or multiscale guards and uncertainty-aware tolerances made selection usable. Compute uncertainty at the level of the claim, preserving event clustering and paired comparisons where appropriate. A small global gain across seeds and an equally sized fluctuation in one sparse bin are not equally strong evidence.

G Model Names and Architectures

G.1 How to read a model name

A model name lists its inputs and architecture, then its training objective and classifier (Table 7). Two examples:

- "E1+V4+F, IF": an EM2 branch of type E1, a core-tensor backbone of type V4 and the $\mathrm{Frac}_{\mathrm{EM2}}$ scalar, joined in the dense head and trained with inverse-frequency weighted BCE. The classifier is NN-only (the default).
- "MLP, CT+F, FT": the $[32, 16]$ MLP on the 45 core cells plus $\mathrm{Frac}_{\mathrm{EM2}}$, pretrained with weighted BCE and then fine-tuned with the constrained objective.

A V-backbone implicitly consumes CT; an E-branch implicitly consumes the fine EM2 image. A suffix $+$F adds the engineered scalar, not another image. Flat CT or SUM bypasses convolution and enters the head directly. Unless OR is specified, the classifier is NN-only. Thus E3+V2+F, FT means EM2 branch E3, core backbone V2 and Fraction, first trained with IF and then constrained-fine-tuned; it does not mean a power-law run.

G.2 Common interface

All models share one interface: `forward` returns a probability and `forward_logits` returns the pre-sigmoid score. Training, calibration and constrained fine-tuning therefore treat every model in the same way. A configuration without a `model` block reproduces the MLP exactly. Setting `model.name=tensor_cnn` builds a convolutional model from a list of branches. Each branch names a registry feature and a shape, and folds that feature's columns back into a (channels, height, width) image in the same order as the registry. Kernel sizes, channel counts, strides, pooling, extra branches and skip connections are all configuration fields, so a new architecture is a new configuration rather than new code.

G.3 MLP

The MLP is $d \to 32 \to 16 \to 1$ with ReLU activations and a sigmoid output, and has $32d + 577$ parameters. For example, it has $2{,}049$ parameters on CT+F ($d = 46$), $2{,}145$ on CP+CT and $1{,}089$ on SUM+F. Kernan and Areva Cohen [5] also swept the hidden layers $[48]$, $[32, 32]$, $[64, 32]$, $[32, 16, 8]$, $[64, 32, 16]$, $[64, 32, 16, 8]$, $[64, 32, 16, 8, 8]$ and $[512, 256, 128, 64, 32]$ over seven feature sets, and found no systematic change in the turn-on curve. The largest network was the weakest on the largest input.

Table 7: Naming key for inputs, architectures, objectives and classifiers.

Name	Meaning
Inputs	
CT	the 45 raw cell energies of the $5 \times 3 \times 3$ core tensor (Appendix B)
SUM	15 per-layer summaries: energy sum, core dominance and sparsity of each of the five layers
CP	"core physics": BDT score, $p_{\mathrm{T}}^{\mathrm{TOB}}$, and η, ϕ relative to the leading TOB
F	the EM2 compactness $\mathrm{Frac}_{\mathrm{EM2}}$ (Equation (3))
EM2	the 12×12 EM2 image; read only by an E-branch
flat CT / SUM	the cells or summaries passed directly to the dense head, without convolution
Architectures	
MLP	fully connected network $d \to 32 \to 16 \to 1$
V1	core-tensor layer mixing: 1×1 convolution across five input layers, 12 filters
V2	core-tensor spatial processing: 2×2 convolution across five layers, 12 filters
V3	serial core backbone: V1 followed by a 2×2 convolution on its feature maps
V4	parallel core backbone: V1 and V2 outputs concatenated (Appendix G.4)
E1	EM2: 3×3 convolution, 4 filters, then 2×2 max pooling
E2	EM2: 3×3 convolution, 8 filters, pooling, then a second 3×3 convolution
E3	EM2: as E2, but stride 2 replaces first-stage pooling (Appendix G.5)
Ex+Vy	dual-branch model: E-branch and V-backbone concatenated in one head
Ex + flat CT	EM2-only model: an E-branch plus plain core-cell scalars (Appendix G.7)
Training objective	
BCE	plain binary cross-entropy
IF	inverse-frequency BCE: signal weights from empirical training-bin counts, with clipping (Appendix E)
p	power-law BCE: weights proportional to clamped t^{-p}; $p < 0$ emphasises energetic signal. Not the IF formula
FT	constrained fine-tuning of a pretrained model; an unqualified CNN FT starts from IF
BCE+FT / $p=-1$+FT	FT after plain BCE / power-law BCE with $p = -1$, respectively
rank control	earlier rank-calibrated soft-objective FT control, not the pairwise tail-ranking variant
Classifier	
NN-only	default: an object is accepted when $s_{e,i} \geq t$
OR	TOB–NN OR classifier (4)

G.4 Core-tensor backbones V1–V4

The core tensor is a stack of five 3×3 images, one per calorimeter layer (Figure 7a). A convolution slides one small set of weights, the *kernel*, over the image and applies the same weights at every position. This builds in two assumptions an MLP must learn from data: nearby cells are related, and the same pattern means the same thing wherever it appears. On a 3×3 grid a 3×3 kernel fits in only one position, so it is no different from a dense layer. Only two kernel shapes give real weight sharing:

- **V1, layer mixing ($1 \times 1 \times 5$).** At each of the nine cells, the kernel reads the five energies of that cell, one per layer from EM0 to HAD, and forms a weighted sum. Twelve such filters give twelve numbers per cell. Each filter is a learned description of the *longitudinal profile*, i.e. how the energy is shared between the layers, and the same filter is applied at every cell. The output is $12 \times 3 \times 3 = 108$ values from 72 weights.
- **V2, spatial ($2 \times 2 \times 5$).** The kernel reads a 2×2 block of cells across all five layers (20 energies) and fits at four positions. Its twelve filters learn local spatial patterns, such as "energy in the centre but not in its neighbour", in all layers at once. The output is $12 \times 2 \times 2 = 48$ values from 252 weights.
- **V3, serial.** V1 first converts each cell into twelve profile values. A 2×2 convolution over these twelve channels then looks for spatial patterns in the learned profiles. The output is 48 values.
- **V4, parallel.** V1 and V2 run side by side on the same input, and their outputs are concatenated ($108 + 48 = 156$ values). Each mechanism can specialise without passing through the other.

The window is centred on the TOB seed, so the absolute position of a pattern is informative: a large central cell means something different from a large corner cell. We therefore never apply global pooling; the feature maps are flattened and fed to the dense head together with any scalar inputs.

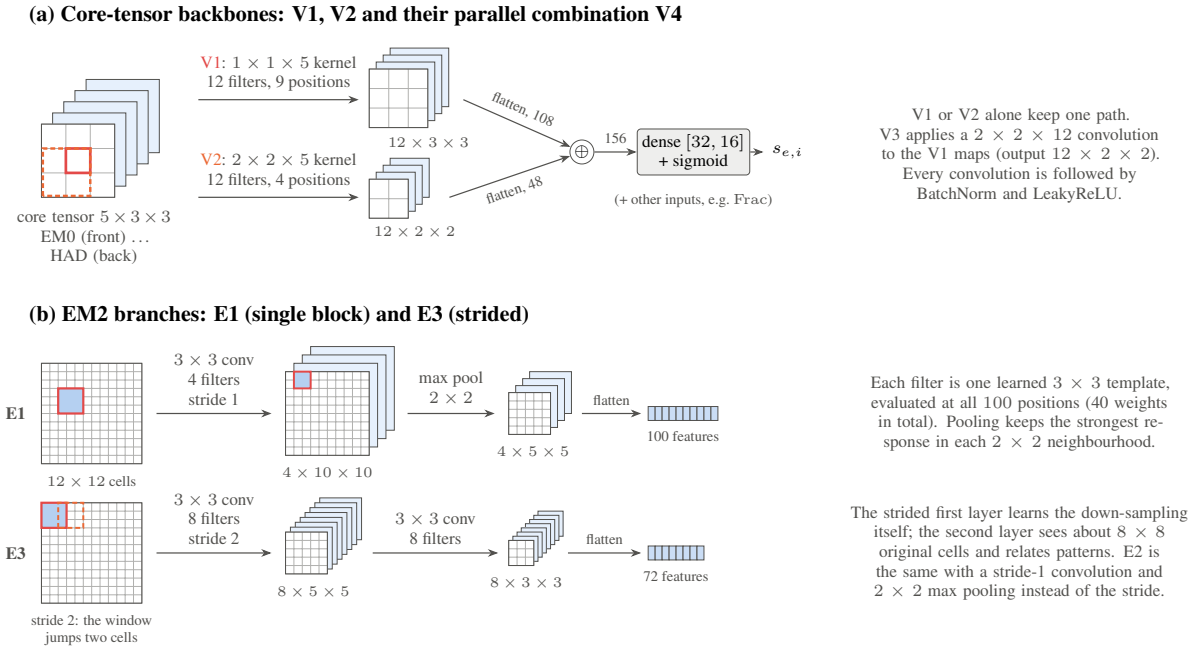


Figure 7: Convolutional building blocks. (a) The core-tensor backbones: V1 applies a $1 \times 1 \times 5$ kernel (red: one cell through all five layers) at nine positions; V2 applies a $2 \times 2 \times 5$ kernel (orange dashed) at four positions; V4 runs both and concatenates their flattened maps before the dense head. (b) The EM2 branches: E1 (one 3×3 convolution with four filters and 2×2 max pooling) and E3 (a stride-2 convolution followed by a second convolution).

G.5 EM2 branches E1–E3

The 12×12 EM2 image is large enough for standard convolution blocks (Figure 7b).

- **E1, single block.** A 3×3 kernel slides over the image with stride 1, which gives $10 \times 10 = 100$ positions. Four filters with 9 weights and a bias each (40 parameters) act as four learned templates, such as an isolated peak or an energy gradient. Their responses form four 10×10 maps. A 2×2 max pooling keeps only the strongest response in each 2×2 neighbourhood, so the maps shrink to $4 \times 5 \times 5 = 100$ values. The pooling means "this template fired this strongly somewhere here". Max is used rather than average because it preserves sharp energy peaks.
- **E2, two blocks.** The same block with eight filters ($8 \times 5 \times 5$) is followed by a second 3×3 convolution on the pooled maps ($8 \times 3 \times 3 = 72$ values). Each pooled value already summarises about 4×4 cells, so a second-layer unit sees about 8×8 cells of the original image. It can therefore learn *relations* between patterns, for example two separated clusters or a core with sideways leakage.
- **E3, strided.** As E2, but the first convolution moves two cells at a time (stride 2). It thus produces $8 \times 5 \times 5$ directly, without pooling. The network then learns which information the down-sampling keeps, instead of relying on the fixed max rule [10].

G.6 Engineered normalisation beside a learned EM2 branch

The question was whether an E-branch subsumes $\mathrm{Frac}_{\mathrm{EM2}}$, not whether the scalar introduces an independent measurement. Its numerator is an all-ones 3×3 filter followed by a maximum over windows, so much of its structure resembles convolution and pooling. If the learned branch recovered the full operation, explicitly supplying the scalar would be redundant. The corresponding ablation is reported in Appendix I.5. A plausible explanation is the final division by the total EM2 energy: this global normalisation makes the ratio invariant to a common energy rescaling, whereas a compact local convolution–pooling stack does not readily synthesise it. The scalar supplies that operation directly and leaves the branch free to learn complementary local patterns. Thus, the ablation establishes a practical benefit for the tested networks, not information absent from the original image or an impossibility for larger networks.

G.7 Dual-branch and EM2-only models

A dual-branch model Ex+Vy feeds the core tensor to a V-backbone and the EM2 image to an E-branch. It concatenates both flattened outputs, optionally together with $\mathrm{Frac}_{\mathrm{EM2}}$, into one $[32, 16]$ head. The EM2-only model removes the V-backbone. The 45 core cells (flat CT) or the 15 summaries (flat SUM) enter the head as plain scalars next to the E-branch output. This keeps the longitudinal information of the core tensor, which a dense layer reads well, and removes the shared-weight convolution on the coarse 3×3 window. The motivation was twofold: the EM2 image seemed to carry most of the discriminating information, and the smaller model fits the hardware budget better (Table 8).

Exact model sizes are collected in Table 8; Appendix I.5 measures the performance cost of replacing the V-backbone by flat CT or SUM. "EM2-only" therefore describes the sole *convolutional* branch: it does not mean that the model discards the other layers. The experiment separates the value of longitudinal measurements from the cost of processing them with convolutions.

G.8 Hardware scope

The indicative specification allowed about $4{,}000$ 8-bit weights and restricted available inputs and operations. These requirements changed during the project: an early 49-input design was later restricted to features derived from the 3×3 layers [5]. We therefore report candidates at several sizes rather than claim one fixed deployment optimum: roughly 2k parameters for an MLP, 4–5k for EM2-only convolutional models and 9k for dual-branch models. Trainable counts include BatchNorm parameters, which can be folded into the convolutions at inference. Counts describe model complexity, not measured FPGA latency or resource use; quantisation, operation support and input availability must also be validated before deployment.

G.9 Input scaling and stability

The raw cell energies span about five orders of magnitude. We first compressed the convolutional inputs logarithmically ($E \mapsto \log(1 + E)$, then standardisation), reasoning that per-column standardisation gives every position its own scale and could defeat weight sharing. A controlled test refuted this. The flat MLP retrained on the compressed input lost 3.8 points of global efficiency relative to the same MLP on standardised input, and 8.3 points in the 25–$40\,\mathrm{GeV}$ band. The CNNs were far more robust (about -0.9 points), presumably because BatchNorm re-centres every channel. The amplitude scale is therefore discriminating information, not a nuisance, and all models use per-column standardised energies. A plain-ReLU CNN collapsed to a constant output within the first epoch on the full, imbalanced data: the first background-dominated updates made all pre-activations negative, and the dead units never recovered. The `conv`→BatchNorm→LeakyReLU stack prevents this. BatchNorm folds into the convolution at inference, so it adds nothing to the deployed cost.

G.10 Parameter counts

Table 8: Trainable parameters, including BatchNorm (which folds into the convolutions at inference). Names: Table 7. The head is always $[32, 16]$. "+F" adds the $\mathrm{Frac}_{\mathrm{EM2}}$ scalar to the head input.

Model	Params	Model	Params	Model	Params
MLP, CT	2,017	V1 (CT)	4,129	E1+V1+F	7,409
MLP, CT+F	2,049	V2 (CT)	2,389	E1+V2+F	5,669
MLP, CP+CT	2,145	V3 (CT)	2,821	E1+V4+F	9,221
MLP, SUM+F	1,089	V4 (CT)	5,941	E2/E3+V1+F	7,161
E1, flat CT+F	5,297	V2+F	2,421	E2/E3+V2+F	5,421
E2/E3, flat CT+F	5,049	V4+F	5,973	E2/E3+V4+F	8,973
E2/E3, SUM+F	4,089	E1, SUM+F	4,337		

The dense head holds most of the parameters (82–98% in the dual-branch models). Pruning it and quantisation-aware training are therefore the natural first steps towards the FPGA budget.

H Training and Selection Details

Hyperparameter sweeps. The optimiser settings were fixed by sweeping the learning rate over $\{10^{-4}, 10^{-3}, 5 \cdot 10^{-3}\}$ and the batch size over $\{256, 1024, 4096\}$. Each configuration was trained with the same three seeds and

Table 9: Hyperparameters. Every value is a field of the experiment configuration.

Setting	Value
BCE and weighted-BCE training (MLP and CNN)	
Optimiser	Adam, learning rate 10^{-3}
Batch size, epochs	256 objects, 20 epochs
Input preprocessing	per-column standardisation, fitted on the training split
Seeds (and event splits)	42, 123, 456
Checkpoint	target-FPR on validation at $\varepsilon_{\mathrm{B}} = 0.005$ (primary); minimum validation BCE (reference)
Signal weights	plain; $\alpha \in \{0, 1, 2, 4\}$; inverse frequency (25–100 GeV, 5 GeV bins, clip $[0.2, 5]$); power law (t clamped to $[10, 200]$ GeV)
OR classifier	
TOB budget grid	$b \in \{0, 0.0005, \ldots, 0.005\}$, two-fold validation cross-fitting
Objective windows	5 GeV from 25 to 100 GeV, equal weights
Guards	per window / pooled $[60, 120)$ / multiscale ($z = 1$, $\delta_{\mathrm{phys}} = 0.005$)
Constrained fine-tuning	
Initialisation	best weighted-BCE checkpoint of the same representation
Optimiser	Adam, learning rate 10^{-4}, 20 epochs, batches of 2048 complete events
Training split	primal 70% / constraint 30% of training events (stratified)
Temperature	cosine schedule $0.10 \to 0.02$
Proxy threshold	batch-rank background quantile at the event target
Dual ascent	$\eta_{\mathrm{FPR}} = 1$, $\eta_k = 50$, $\lambda_{\max} = 25$, all λ start at 0
Regions	objective $[25, 32)$, $[32, 40)$, $[40, 60)$; constraints add $[60, 120)$
Margins	$m = (0.0025, 0.0025, 0.0025, 0)$, $\rho = (0.0025, 0.0025, 0.0025, 0.005)$
Selection	validation cross-fitting, one-sided 95% guards with $\delta_{\mathrm{phys}} = 0.005$
Deployment	confidence-safe Clopper–Pearson target (95%)

compared through an inverse Fermi–Dirac fit of the three-seed mean turn-on curve, $\epsilon(p) = \epsilon_{\max}/\left(1 + e^{-k(p-p_{50})}\right)$, with plateau $\epsilon_{\max}$, slope k and midpoint p_{50}. A lower midpoint means an earlier turn-on. Table 10 reproduces the sweep of the earlier study [5], which used the baseline MLP on its 49-input and 193-input configurations. The combination of learning rate 10^{-3} and batch size 256 gave the lowest midpoint within a spread of about one point in plateau. We repeated the same grid on our own models and reached the same choice, so all runs use these values. All runs train for 20 epochs. Because the target-FPR checkpoint is selected after every epoch, a longer run can only add candidate checkpoints; it cannot replace a good early epoch.

Table 10: Learning-rate and batch-size sweep (inverse Fermi–Dirac fit parameters of the three-seed mean turn-on curve; midpoint in GeV; values from [5]). The selected values are marked.

Learning rate	Midpoint	Slope	Plateau	Batch size	Midpoint	Slope	Plateau
10^{-4}	30.67	0.159	0.869	256 (selected)	29.33	0.153	0.863
10^{-3} (selected)	29.14	0.154	0.861	1024	29.66	0.155	0.865
$5 \cdot 10^{-3}$	29.31	0.151	0.866	4096	30.12	0.157	0.868

The hidden-layer sweep of the MLP is summarised in Appendix G.

Target-FPR checkpoint selection. Minimum validation BCE averages over all objects, including abundant easy examples far from the trigger boundary. A lower BCE need not improve the ordering of scores near the background tail that determines the 0.5% event cut. We therefore select by validation performance at the operating point and retain the minimum-BCE checkpoint only as a reference. Depending on the experiment, the configured criterion is signal efficiency or the trigger objective with its feasibility requirements; constrained training uses the cross-fitted protocol in Appendix F. The test set never chooses features, losses, budgets, architectures or checkpoints.

At the end of every epoch the network threshold is calibrated on the validation background, so that $\widehat{\mathrm{FPR}}_{\mathrm{val}} \leq 0.005$ with the event rule (1), or with the configured OR classifier including its budget search. The signal efficiency is then measured on the validation truth τ objects. The epoch with the highest efficiency is kept, with ties broken by achieved fake rate and then by validation BCE. The two criteria are never merged into one score. Both checkpoints are saved, and the configuration names the primary one.

Reproducibility. Seeds are reset before every model is built. Data alignment, event splits and raw features may be cached across the runs of one sweep, but normalisation statistics, models, optimiser states and all validation-derived choices are recomputed for each run. A cache hit changes neither the random state nor the results; regression tests check this equivalence. Experiments ran on the Tel Aviv University CS GPU cluster (Slurm). The code, all experiment configurations and a demo are available at https://github.com/Maxgolu/Tau-Trigger-NN.

I Detailed Results

I.1 How the numbers are computed

Our result tables are recomputed from stored test runs; the separately supplied prior-work reference is described in Appendix I.2. For each seed, the model and the baseline are calibrated on the test background to the same 0.5% event-level target, and the achieved rates are recorded (Appendix C). A band efficiency pools the $2.5\,\mathrm{GeV}$ bins of that band, each weighted by its number of test τ objects, so it equals the fraction of all τ objects of the band that are accepted. The table entry is the mean over the three seeds, and $\pm$ is the standard error of that mean. A configuration enters the lists only when all three seeds exist and the configurations differ only in the seed. The efficiency at the threshold calibrated on validation is stored for every run as a separate check of generalisation.

I.2 Historical reference

The previous researchers supplied the five band efficiencies in Table 1: 12.59, 60.07, 86.44, 90.90 and 91.63%, with a confirmed global efficiency of 62.53%. These are externally supplied results rather than measurements recomputed in this study. The prior row remains separate from our fixed-protocol rankings and red baseline-deficit markings because its evaluation protocol differs.

Figure 4b retains the published Fermi–Dirac fits (NN midpoint $26.8741\,\mathrm{GeV}$, slope $0.1830\,\mathrm{GeV}^{-1}$ and plateau 0.9157). Band averages are not efficiencies at bin centres; we neither reconstruct a new fine-grained curve from the five supplied means nor adjust the means to match that fit.

I.3 Benchmark trade-offs and seed-level behaviour

All five models turn on much earlier than the baseline (Figure 4a). They reach 50% efficiency at 26.5–28.1 GeV instead of $35\,\mathrm{GeV}$ and 90% at 47–50 GeV instead of $55\,\mathrm{GeV}$. They gain 25–31 points in the 25–40 GeV band and 10.1–12.2 points globally. They differ in the last bins. Model (1) is the most efficient model above the baseline everywhere, and it pays 1.6 global points for this relative to model (2). It is trained with plain BCE and uses the OR classifier, whose TOB branch recovers the energetic τ objects that plain BCE neglects. The criterion holds for the three-seed mean: in two seeds the validation search gave the TOB branch 1.5×10^{-3} of the budget and the model saturates at 99.9%, while the third seed chose 0.5×10^{-3} and falls below the baseline above $60\,\mathrm{GeV}$ (0.1 points at 60–100 GeV and 0.8 at 100–120 GeV). Models (2)–(4) give up 0.5–1.1 points at 100–120 GeV. This band holds only about 750 τ objects per seed, so one point corresponds to seven or eight objects. The MLP (4) is only 1.2 points below the best model at less than a quarter of its size. Model (5), V3+F with IF weighting and the OR, is the compact counterpart of model (1): with 2.9K parameters it stays above the baseline everywhere at a cost of 0.5 global points relative to model (1). Its validation-selected TOB budget was zero in two of the three seeds, so its plateau comes mainly from the network; the serial backbone V3 favoured the plateau over the turn-on in all three CNN sweeps. The same network without the OR also satisfies criterion (5) (67.39%), but beats the baseline at 100–120 GeV by only 0.03 points; it is the option for a firmware without an OR. The best model of the previous work (Figure 4b) also turned on earlier than its baseline, but it fell below it at $39\,\mathrm{GeV}$ and saturated at 91.6%. Our benchmark saturation values are within about one point of our baseline; the prior-work comparison remains qualitative because its protocol differs.

I.4 Weighting and the value of the OR branch

The hypothesis of Section 4.2 concerns relative emphasis on energetic signal, not one universally optimal weighting formula. The initial turn-on-emphasis screen did not consistently help; the later power-law screen favoured negative p, and IF improved two of the three input representations in the initial matched comparisons. These observations support the proposed benefit of learning from clearer energetic showers, but do not establish that mechanism causally or imply that all models benefit. Both IF and power-law experiments are documented in Appendix E.

Table 11 asks whether signal weighting reduces the need for a deterministic TOB branch. Read across each row while holding inputs fixed: compare BCE+OR with IF+OR for weighting, then IF+OR with IF NN-only for the classifier.

Missing entries mean that a complete four-way comparison is not available; the SUM+F row, in particular, must not be read as a matched BCE-to-IF improvement. Parentheses expose saturation costs that global efficiency alone hides. Model-name definitions are in Table 7.

Table 11: Weighting and OR comparisons on fixed MLP inputs. Entries are global efficiency [%], with 100–120 GeV efficiency in parentheses. OR entries additionally give mean TOB budget $\bar{b}^\star$ in units of 10^{-3}. IF is inverse frequency, not t^{-p} (Table 7 and Appendix E).

	BCE		IF	
Input	NN-only	OR ($\bar{b}^\star$)	OR ($\bar{b}^\star$)	NN-only
CT	—	64.53 (99.82) 2.5	65.92 (99.37) 0.5	67.03 (99.02)
CT+F	—	63.11 (99.73) 3.3	66.97 (99.69) 1.2	**68.40** (98.53)
SUM+F	67.82 (96.64)	—	66.40 (99.61) 0.5	67.51 (98.62)
Baseline	57.37 (99.34)			

Trained with plain BCE, the network neglected energetic τ objects: the best NN-only models lost 2–3 points to the baseline at 100–120 GeV (Table 11). The OR classifier repaired the plateau, but its TOB branch took half or more of the fake-rate budget ($\bar{b}^\star = 2.5$–3.3×10^{-3}) and the global efficiency fell to 63–65%. Weighting the signal towards high energy changed the picture. With inverse-frequency weights the selected budgets fell to 0.5–1.2×10^{-3}, and the NN-only models reached 98.5–99.0% at 100–120 GeV on their own. Removing the OR then raised the global efficiency by 1.1–1.4 points. Figure 5a shows why: after reweighting, the OR still adds up to 1.2 points above 60 GeV, but it pays for them with 2–4 points below 40 GeV, where most τ objects are. The same trade appeared in the CNNs: on the strongest inputs with plain BCE, the OR cost 1.1–2.5 global points. The OR is still the most reliable way to exceed the baseline at 100–120 GeV, which is why the model of criterion (1) uses it; one NN-only model, V3+F, also exceeds it, by 0.03 points.

Power-law weighting is a separate tested control, not hidden under the IF label. For CT+F with the OR, $p = -1$ gave 66.66% globally and 99.73% at 100–120 GeV, versus 66.97% and 99.69% for IF. Under NN-only, the corresponding pairs were 67.81/99.33% and 68.40/98.53%. Both high-energy-emphasising schemes were useful, but they selected different turn-on/plateau trade-offs.

I.5 Spatial representation, Fraction and compression

A convolution on the core tensor alone did not beat the flat MLP (V4: 66.68 vs. 67.03). Adding the scalar $\mathrm{Frac}_{\mathrm{EM2}}$ raised V4 by 1.6 points, almost entirely in the turn-on: +3.3 points at 10–25 GeV and +2.8 at 25–40 GeV (Figure 5b). A learned EM2 branch in place of $\mathrm{Frac}_{\mathrm{EM2}}$ gained slightly more (E1+V4, 68.67). Adding $\mathrm{Frac}_{\mathrm{EM2}}$ back next to the branch still helped (69.15 ± 0.14), again in the soft region (+1.4 and +1.0 points), supporting a practical benefit from supplying the ratio explicitly, rather than proving that convolutional networks cannot learn it. The core-tensor convolution itself then turned out to be dispensable. The EM2-only model, with the core cells as plain scalars, reached 68.88 ± 0.19 with 43% fewer parameters than E1+V4+F. Averaged over the four backbones, $\mathrm{Frac}_{\mathrm{EM2}}$ added 1.35 points, the 15 layer summaries only 0.25, and the core-physics variables lowered the efficiency on every backbone, by about one point (by 0.2–0.3 on the MLP), consistent with the energy-shortcut explanation in Section 4.1.

The error-set comparison provides a complementary observation: in 25–40 GeV, at least one of the four core CNNs accepted 1,354 taus missed by the flat MLP, while 201 were accepted by the MLP but missed by all four CNNs. This comparison uses the union of four CNN acceptances, not a single CNN at its operating point. It supports complementary pattern sensitivity despite no global core-CNN advantage; it does not identify individual learned filters or prove universal CNN superiority. The architectural explanation for combining learned local patterns with explicit normalisation is given in Appendix G.6.

I.6 Capacity and shared errors

Table 12 asks a different question from Table 11: does changing architecture or size help when CT, IF and the NN-only decision are fixed? The upper block is the controlled comparison. The lower block is contextual evidence from the earlier MLP sweep, expressed in fitted midpoint and plateau rather than our global metric; the two blocks should not be compared numerically.

The four core-tensor backbones span a factor 2.5 in size and four different inductive biases, yet all land within 0.3 points of each other and below the flat MLP (Table 12). They also fail on the same objects: the sets of missed τ

Table 12: Capacity study. Upper block: global efficiencies for our CT-only, IF-trained, NN-only models. Lower block: prior-work fitted midpoint [GeV] and plateau, averaged over seven input sets [5]. Model-name key: Table 7.

Model (ours, CT)	Params	Global
MLP $[32, 16]$	2.0K	67.03 ± 0.36
V1	4.1K	66.70 ± 0.28
V2	2.4K	66.44 ± 0.21
V3	2.8K	66.49 ± 0.38
V4	5.9K	66.68 ± 0.22
MLP sweep [5]	**Midpoint**	**Plateau**
$[32, 16]$	28.0	0.908
$[512, 256, 128, 64, 32]$	28.0	0.880

objects overlap with a Jaccard index of up to 0.89, and 96% of the commonly missed τ objects in 25–40 GeV are missed by the flat MLP as well. The earlier MLP sweep [5] shows the same from the other side: growing the network from $[32, 16]$ to $[512, 256, 128, 64, 32]$ left the midpoint unchanged and lowered the fitted plateau from 0.908 to 0.880. Among the EM2 branches, the deeper E2 and E3 gave no systematic gain over the single-block E1 (on V4+F: 69.15 for E1 against 68.72 and 68.48). Together, these controlled comparisons suggest that input discrimination, rather than capacity, was the main limitation in the tested model families.

For scale, the earlier $[512, 256, 128, 64, 32]$ dense architecture has $198{,}145$ parameters on 45 CT inputs, versus $2{,}017$ for $[32, 16]$: about 98 times as many. This is an architectural size calculation; the earlier sweep's reported fit metrics average over seven input sets, not a matched CT-only comparison against our final models.

A plausible explanation for the limited return from E2/E3 is that these small, seed-centred images can be classified largely from local patterns plus their positions. E1's flattened maps and dense head already combine information across the image, so extra convolutional depth need not add useful discrimination. Further down-sampling can also discard fine spatial detail. These are explanations consistent with the ablations, not established causes: the runs do not isolate information loss, optimisation difficulty or overfitting.

I.7 Fine-tuning as an energy-profile control

Table 13 isolates fine-tuning rather than comparing unrelated best models: each pair has the same architecture and inputs, with its pretrained checkpoint followed by FT. The MLP starts from power-law BCE with $p = -1$; the CNN starts from plain BCE. The MLP and CNN objectives use different energy-region settings, explicitly recorded in Appendix F.

Table 13: Matched before/after constrained fine-tuning (FT). Efficiencies [%] by true-p_{T} band [GeV]; red denotes a deficit to the baseline. Names and pretraining suffixes: Table 7.

Model	10–25	25–40	40–60	60–100	100–120	Global	Params
MLP, CT+F, $p=-1$	12.31	62.68	90.44	97.90	99.33	67.81±0.35	2.0K
MLP, CT+F, $p=-1$+FT	13.73	63.84	90.50	97.52	99.01	68.34±0.42	2.0K
E1+V4+F, BCE	19.05	66.91	89.49	95.28	96.89	69.53±0.23	9.2K
E1+V4+F, BCE+FT	17.45	66.39	90.54	96.75	98.28	69.61±0.29	9.2K

The constrained fine-tune moved efficiency towards the weak end of its starting point, and usually changed global efficiency by around half a point or less (Table 13). The $p = -1$ MLP was strong at saturation, and the fine-tune moved 1.2–1.4 points into the 10–40 GeV bands while giving up 0.3–0.4 above 60 GeV. The BCE-trained dual-branch CNN was strong in the soft band, and the fine-tune moved efficiency from there into saturation. Averaged over the nine E+V+F pairs, its effect on the global efficiency was -0.05 points (from -0.46 to $+0.29$). The best weighting also depends on the input: on the strongest input set, plain BCE beat inverse frequency (69.53 vs. 69.15). Finally, across the evaluated configurations no model exceeded 19.1% at 10–25 GeV.

I.8 Four lists, four questions

Each list first applies a hard filter that encodes one question, and then ranks the remaining models by global efficiency, the single number that summarises the gain over the whole spectrum. Filtering first and ranking second avoids weighting criteria that have no common unit, such as a point of saturation and a point of turn-on.

Balanced leaders (Table 14). Models with at least 98.5% at 100–120 GeV, excluding core-tensor convolution models of 6K parameters or more. The question: which models are strong at both ends of the spectrum and plausibly deployable?

Plateau experts (Table 15). Models with at least 99% at 100–120 GeV. Saturation is the strictest requirement of the trigger, and this list shows what reaching it costs.

Most promising (Table 16). The ten highest global efficiencies, with no other condition. Models above 6K parameters are marked †: before deployment they need compression, pruning, quantisation or a smaller equivalent configuration. Several need only a targeted repair of one energy region, usually saturation.

Within budget (Table 17). The five highest global efficiencies among models of at most 5K parameters, the budget closest to the specification we received.

The five benchmark models of Section 5.1 answer related but stricter questions and are chosen by the same filter-then-rank rule.

Because every value is recomputed, the lists differ in places from those of the project report, whose saturation-band values for the CNNs were 0.2–0.7 points too low. The most visible change is in the plateau list: NN-only models also reach 99% at 100–120 GeV, so a high plateau does not require the OR. Exceeding the baseline there did, with one marginal exception (NN-only V3+F, +0.03 points). The criterion-(1) model also changes: the plain-BCE OR grid, whose saturation the report underestimated, contains the most efficient model that beats the baseline in every band. The lists are built from the 104 three-seed configurations we re-evaluated. They cover the complete 36-cell grid of dual-branch models, every V-backbone model with or without $\mathrm{Frac}_{\mathrm{EM2}}$, every EM2-only model, and every model that the project report ranks among its best. The runs not re-evaluated are a few early OR and weighting variants, the V-backbones with CP or SUM added, and the reproduction of the previous work; by their reported global efficiencies (at most 67.8%) and saturation values, none of them could enter any list.

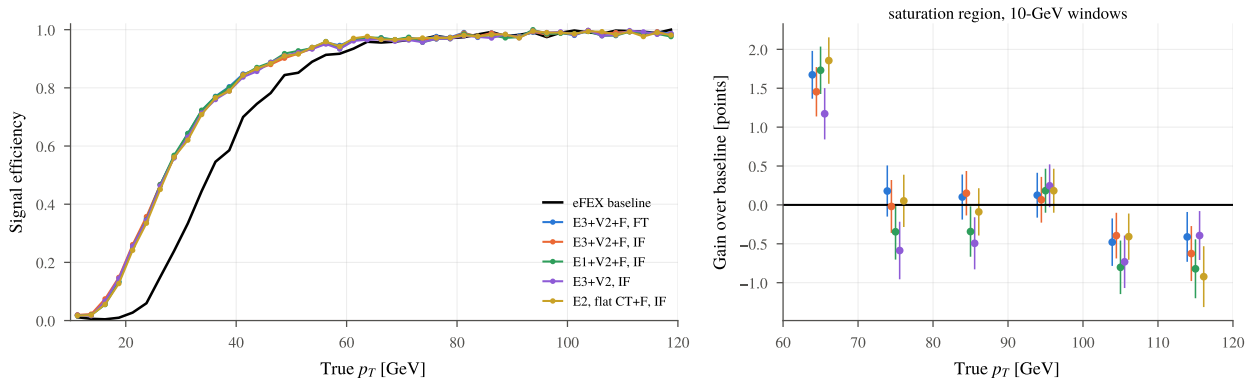


Figure 8: Balanced leaders. Model-name key: Table 7. Left: turn-on curves. Right: gain over the baseline in 10 GeV windows of the saturation region; error bars are the binomial uncertainty of the model efficiency.

Table 14: Balanced leaders: the five best global efficiencies among models with at least 98.5% at 100–120 GeV, excluding core-tensor convolution models of 6K parameters or more. Model-name key: Table 7.

Model	10–25	25–40	40–60	60–100	100–120	Global	Params
E3+V2+F, FT	16.93	64.31	90.08	97.56	98.89	69.08±0.35	5.4K
E3+V2+F, IF	17.73	63.81	89.78	97.43	98.84	69.05±0.25	5.4K
E1+V2+F, IF	16.65	64.40	90.29	97.36	98.53	68.96±0.12	5.7K
E3+V2, IF	17.08	63.48	89.69	97.08	98.75	68.68±0.44	5.4K
E2, flat CT+F, IF	16.04	63.11	90.05	97.56	98.70	68.55±0.53	5.0K
eFEX baseline	2.28	35.85	81.60	96.89	99.34	57.37	—

The balanced list is led by the strided E3+V2+F pair, with and without fine-tuning (69.08 and 69.05); the first is the model of criterion (3). All five balanced leaders read the EM2 image and have 5.0–5.7K parameters.

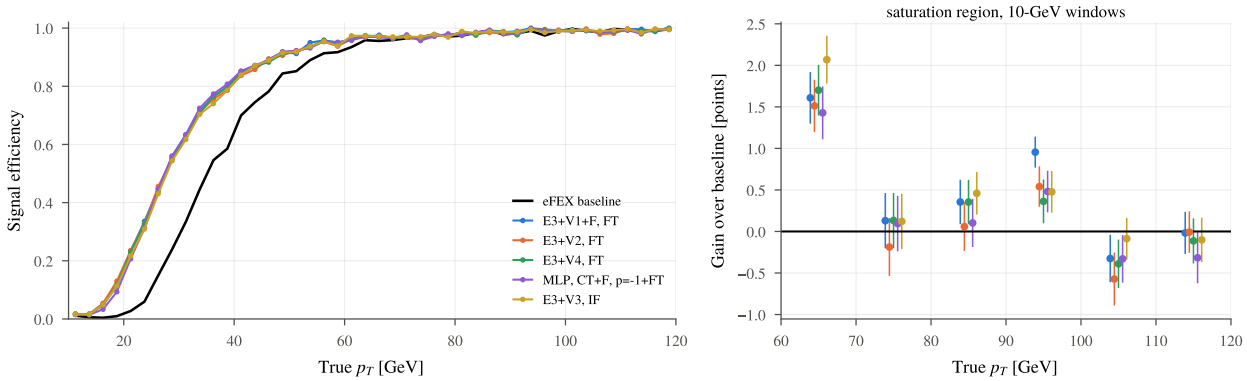


Figure 9: Plateau experts. Model-name key: Table 7. Left: turn-on curves. Right: gain over the baseline in 10 GeV windows of the saturation region; error bars are the binomial uncertainty of the model efficiency.

Table 15: Plateau experts: the five best global efficiencies among models with at least 99% at 100–120 GeV. Model-name key: Table 7.

Model	10–25	25–40	40–60	60–100	100–120	Global	Params
E3+V1+F, FT	15.43	63.37	90.54	97.73	99.15	68.61±0.11	7.2K
E3+V2, FT	15.77	63.06	89.84	97.47	99.01	68.44±0.87	5.4K
E3+V4, FT	15.28	62.92	90.08	97.65	99.07	68.35±0.14	8.9K
MLP, CT+F, $p=-1$+FT	13.73	63.84	90.50	97.52	99.01	68.34±0.42	2.0K
E3+V3, IF	14.53	61.81	90.10	97.82	99.24	67.98±0.18	5.8K
eFEX baseline	2.28	35.85	81.60	96.89	99.34	57.37	—

All plateau experts pay for their saturation in the turn-on: their efficiency at 10–25 GeV is 13.7–15.8%, against 17.5–19% for the two global leaders. All five are NN-only, four of them constrained fine-tunes; the OR model of criterion (1) misses the list by 0.01 points.

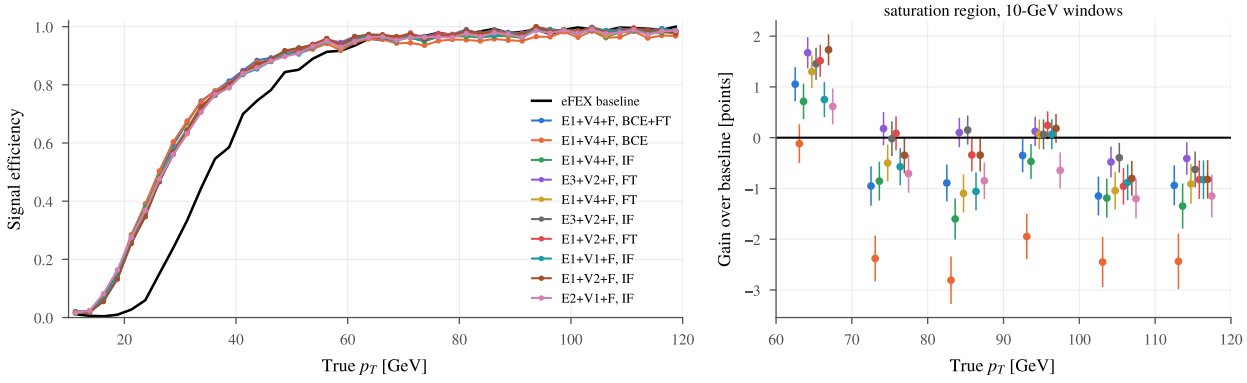


Figure 10: The ten highest global efficiencies. Model-name key: Table 7. Left: turn-on curves. Right: gain over the baseline in 10 GeV windows of the saturation region; error bars are the binomial uncertainty of the model efficiency.

Table 16: The ten highest global efficiencies, with no other condition. †: above 6K parameters. Model-name key: Table 7.

Model	10–25	25–40	40–60	60–100	100–120	Global	Params
E1+V4+F, BCE+FT	17.45	66.39	90.54	96.75	98.28	69.61±0.29	9.2K†
E1+V4+F, BCE	19.05	66.91	89.49	95.28	96.89	69.53±0.23	9.2K†
E1+V4+F, IF	18.51	64.73	89.49	96.49	98.08	69.15±0.14	9.2K†
E3+V2+F, FT	16.93	64.31	90.08	97.56	98.89	69.08±0.35	5.4K
E1+V4+F, FT	17.16	64.78	89.83	96.99	98.35	69.05±0.16	9.2K†
E3+V2+F, IF	17.73	63.81	89.78	97.43	98.84	69.05±0.25	5.4K
E1+V2+F, FT	16.87	64.44	90.28	97.41	98.44	69.05±0.28	5.7K
E1+V1+F, IF	17.84	64.30	89.27	96.79	98.48	68.97±0.30	7.4K†
E1+V2+F, IF	16.65	64.40	90.29	97.36	98.53	68.96±0.12	5.7K
E2+V1+F, IF	18.68	63.78	89.32	96.62	98.16	68.96±0.28	7.2K†
eFEX baseline	2.28	35.85	81.60	96.89	99.34	57.37	—

Every entry of the global top ten reads the EM2 image with a convolutional branch, and six of the ten exceed 6K parameters. Before deployment they need compression, or one of the compact variants below.

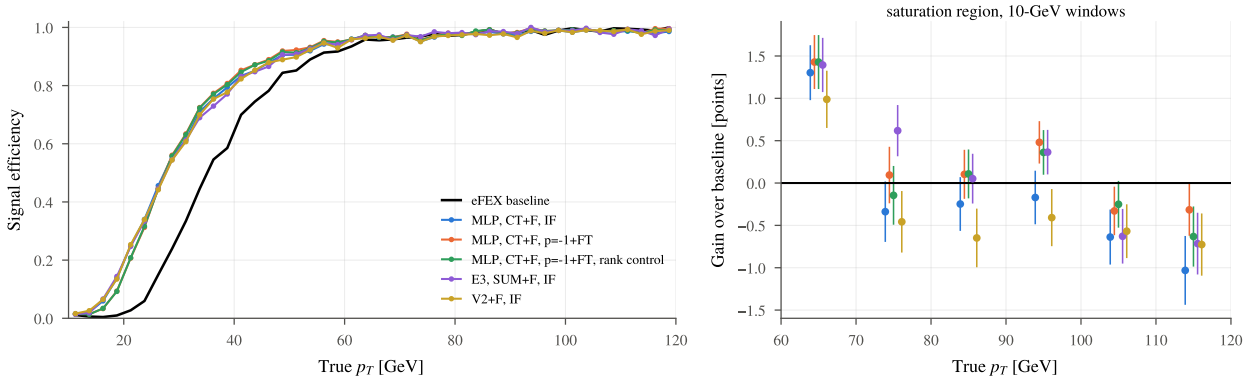


Figure 11: Best models within a 5K budget. Model-name key: Table 7. Left: turn-on curves. Right: gain over the baseline in 10 GeV windows of the saturation region; error bars are the binomial uncertainty of the model efficiency.

Table 17: The five highest global efficiencies among models of at most 5K parameters. Model-name key: Table 7.

Model	10–25	25–40	40–60	60–100	100–120	Global	Params
MLP, CT+F, IF	16.61	62.93	89.00	97.17	98.53	68.40±0.21	2.0K
MLP, CT+F, $p=-1$+FT	13.73	63.84	90.50	97.52	99.01	68.34±0.42	2.0K
MLP, CT+F, $p=-1$+FT, rank control	13.82	63.59	90.11	97.43	98.92	68.22±0.44	2.0K
E3, SUM+F, IF	16.62	61.47	88.90	97.61	98.67	68.08±0.08	4.1K
V2+F, IF	16.79	61.89	88.55	96.90	98.70	68.07±0.30	2.4K
eFEX baseline	2.28	35.85	81.60	96.89	99.34	57.37	—

Within the budget, three 2.0K-parameter MLPs lead: the IF model and two constrained fine-tunes of the $p = -1$ model. At this size the hand-made $\mathrm{Frac}_{\mathrm{EM2}}$ carries the EM2 information more cheaply than a convolutional branch. The EM2-only model with the 15 summaries (4.1K) and V2+F (2.4K) follow, 0.3 points behind.

J Potential Extensions

J.1 Smaller convolutional networks

The strongest global-efficiency models use the fine EM2 input, but their parameter counts exceed those of the compact MLPs. The first step is therefore compression rather than greater depth: reduce channel counts and dense-head widths, then test pruning, quantisation-aware training and knowledge distillation into a student of roughly $2{,}000$ parameters. The dense head contains most parameters and is a natural pruning target. Structured reductions should be evaluated alongside sparse pruning, because fewer nonzero weights do not automatically imply lower hardware cost.

Each candidate should retain the event-level calibration protocol and report the full energy-efficiency profile, not just global efficiency. An FPGA-oriented toolchain such as hls4ml can then assess synthesis feasibility, supported operations, latency and resource use. Quantisation and compression may change score ordering, so calibration and efficiency must be re-evaluated after conversion. The present parameter counts are not hardware measurements, and small accuracy losses cannot be assumed in advance (Appendix G.8).

J.2 Adaptive, representation-aware objectives

The constrained objective used manually specified regions, floors and update rates. A continuation is to derive rewarded regions and allowed deficits from the measured weaknesses of the pretrained model, and scale multiplier updates according to each region's statistical resolution. Region choices must remain tied to physics priorities: automatically optimising only the easiest deficiencies would reproduce the trade-offs the constraints were intended to prevent.

Loss and representation should also be studied jointly. IF helped several compact-input configurations, whereas plain BCE followed by fine-tuning gave the highest global efficiency on the richest tested inputs. A modular objective could adapt to the starting score distribution and representation while preserving the requirements in Appendix F.8. Any data-driven choices must use training or validation data, with a held-out evaluation after the choices are fixed.

J.3 Using the whole event

The pair model sees two TOBs at a time and keeps only the highest pair score. A whole-event model could instead learn from all TOBs together. Because the TOBs have no meaningful order, changing their order should not change the decision. Deep Sets provide one way to build this property into a network [17]; Set Transformers can additionally learn interactions between TOBs using attention [18]. Such models should accept a variable number of TOBs, with masks for absent inputs rather than treating padding as detector information. Varying the number of retained TOBs would measure the value of additional context against latency and resource cost.

Pairwise angular separations, relative energies and correlations among shower shapes can then affect one joint event decision. Define whether the output is an event score or context-aware object scores, and match both approaches at the same event-level background rate. Preserve event-level splits, give each event equal weight during training, and report the event metric as well as compatible object-level efficiencies. The current sample retains at most six TOBs: testing candidates beyond that limit requires a new extraction, not merely changing the model's input size. Since the existing final test has already informed the present study, this comparison requires a new held-out evaluation sample.

J.4 Richer inputs for the low-energy region

Low-energy taus remained difficult across substantially different models, and shared errors suggest that the current representation is a limiting factor, though they do not establish a fundamental information bound (Appendix I.6). Candidate additions include finer granularity in non-EM2 layers, wider calorimeter windows, timing or tracking quantities and neighbouring-object context. These are options to investigate only where detector readout and the intended trigger stage make them available.

The existing 12×12 EM2 image resolves the same angular region as the coarse core tensor; it is not already a wider-window measurement. Separate added spatial coverage from added granularity in controlled comparisons, and assess gains within low-energy bins at the fixed event rate. Any new input must justify its acquisition, preprocessing and hardware cost as well as its predictive benefit.

K Pair Model Details and Reproducibility

K.1 Constructing the TOB pairs

For each event, we keep the TOBs with finite measured $p_{\mathrm{T}}^{\mathrm{TOB}}$. We sort them by decreasing $p_{\mathrm{T}}^{\mathrm{TOB}}$, using the original TOB index to break exact ties, and form every unordered pair. An event with n_e retained TOBs therefore produces $\binom{n_e}{2}$ pairs in a fixed order.

During training, a pair is positive when both TOBs are labelled as tau candidates and negative when neither is. Pairs with one labelled and one unlabelled TOB are not used for training. Two positive labels do not prove that the TOBs match two different generator taus. During validation and testing, every pair is scored because truth labels are not available to the trigger. No truth quantity is a network input.

The final test contains $64{,}286$ events, $361{,}685$ finite TOBs and $866{,}872$ pairs. There are 589 events with fewer than two finite TOBs. These events remain in the efficiency and fake-rate denominators and cannot pass through the neural branch.

K.2 How the final model was selected

The pair study was organised as six questions, with one main change at each stage:

(i) **Does the pair pipeline work?** A model using only the two measured $p_{\mathrm{T}}^{\mathrm{TOB}}$ values checked pair identities, training-only normalisation, event reconstruction, threshold calibration and reproducibility across three seeds. It did not beat the $p_{\mathrm{T}}^{\mathrm{TOB}}$ baseline (`configs/pair_pt_control/`, `results/pt_control/`).

(ii) **Which detector inputs help?** With the model held fixed, we compared raw cells, layer summaries and compact shower features. Repeating the study across three seeds showed that calorimeter information beyond the two measured momenta improved validation efficiency (`results/representation_study/`, `results/representation_reproduction/`).

(iii) **Does the fine EM2 image help?** We added one shared encoder for the two full-granularity EM2 images, followed by the absolute and signed two-channel form used by the final model (`results/high_resolution/`, `results/high_resolution_two_channel/`).

(iv) **How should the model protect high-energy events?** Controlled comparisons motivated the $p = -1$ signal weighting and the direct $p_{\mathrm{T}}^{\mathrm{TOB}}$ branch (`results/pair_or/`, `results/high_resolution_pt_or/`, `results/pt_factorial/`, `results/high_energy_recovery/`).

(v) **Could a simpler explanation account for the gain?** We tested alternative context variables, removed selected inputs, simplified the representation and compared shared encoders. Variants that did not improve the retained model were not promoted (`results/pt_context/`, `results/source_ablation/`, `results/clean_representation/`, `results/shared_encoder/`).

(vi) **Which pairs and events should determine the loss?** The final validation study replaced the four-leading-TOB limit with all finite pairs and gave every event equal weight, regardless of its number of pairs. This fixed the final configuration in `configs/pair_final_all_finite_equal_event/` before the test in `results/final_test/`.

All intermediate percentages come from validation. Only `results/final_test/` contains the final test result. The detailed presentation, `presentation/pair_model_research_history.pptx`, preserves the full experiment sequence and its references.

K.3 Training and model selection

The model is trained for 20 epochs with Adam, learning rate 10^{-3}, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, no weight decay and no AMSGrad. Seeds 42, 123 and 456 use loader seed $424200 + \mathrm{seed}$. Each update contains all pairs from at most 256 shuffled events and uses the loss in (9). We save one checkpoint per epoch. Validation data select one epoch and one OR threshold pair for each seed. Normalisation values inherited from the parent model are checked by hash and are never silently refitted.

The checkpoint SHA-256 values are `b6f74de01e81...a5da605`, `0b8147d99078...4fb02a` and `12e193189d58...a16c3`. The repository manifest records the complete hashes, configuration, preprocessing files and source-file hashes.

Table 18: Validation-selected checkpoints and decision thresholds. The direct $p_{\mathrm{T}}^{\mathrm{TOB}}$ branch is effectively disabled for seed 42 by its selected budget.

Seed	Epoch	Neural threshold	TOB budget	C [GeV]
42	17	0.201933	0.000	127.5
123	18	0.351946	0.001	33.8
456	20	0.400091	0.001	33.8

K.4 Final test

All model and threshold choices were fixed before the final test. The test set was not used for retraining, checkpoint selection, threshold changes or recalibration. Signal efficiency uses all $14{,}509$ signal events, including events without a pair; event fake rate uses all $49{,}777$ background events.

Uncertainty is computed from paired event decisions. For each seed, the two-sided 95% interval is obtained from the exact Clopper–Pearson interval [16] for $b/(b+c)$ and transformed to the paired difference $(b-c)/N$. The one-sided exact McNemar p-values for the high-energy region are Holm-adjusted across the three seeds. We report the arithmetic mean and sample standard deviation across seeds. Events are not pooled across seeds, and the three models are not combined into an ensemble.

The public result files intentionally exclude raw detector data, event-level predictions and trained checkpoints. The submitted verifier checks the public configuration, implementation and result-file hashes, recomputes the reported mean and standard deviation, and runs the data-free CPU demo.